Solar-sail ocean drone, 1.5 m

Drawing set for the custom-built parts, with 3D-print files and full-size templates

Sheet	Title
D00	Cover, index and general notes
D01	General arrangement
D02	Hull lines and table of offsets
D03	Interior arrangement and systems
D04	Keel fin and lead bulb
D05	Twin rudders, stocks and steering
D06	Wing sail and tail
D07	Mast step, spar and stern stub
D08	Deck hatch, solar layout, penetrations
D09	Watertight subdivision, foam fill, battery box, labels
D10	Printed parts P01, P02
D11	Printed parts P03, P04
D12	Printed parts P05, P06
D13	Printed parts P07, P08
D14	Printed parts P09, P10
D15	Printed parts P11, P12
D16	Printed parts P13, P14
D17	Printed parts P15, P16
D18	Printed parts P17, P18
D19	Printed parts P19, P20
D20	Printed parts P21, P22
D21	Printed parts P23, P24
D22	Printed parts P25, P26

Part	3D-printed part	Qty	Material / settings
P01	Wind-vane sensor housing	1	ASA or PETG, 100% walls 4
P02	Wind-vane arm with fin and counterweight pocket	1	ASA or PETG
P03	Sensor cap and stub-tube socket	1	ASA or PETG
P04	Magnet holder for vane shaft	1	PETG
P05	Antenna platform clamp	1	ASA
P06	Transom bracket for 20 mm stub tube	1	ASA, 6 walls
P07	Mast bearing housing with catcher lands	1	PETG or ASA, 100% infill
P08	Spar clamp with tail-boom and counterweight-arm sockets	1	ASA or PETG, 100% infill
P09	Tail servo housing (boom end)	1	ASA or PETG
P10	Tail fin, lower and upper halves	1	PETG or ASA, 0.2 mm layers
P11	Rudder tiller arm	2	PETG, 100% infill
P12	Rudder servo cradle (Blue Trail SER-2000)	2	PETG or ASA, 5 walls
P13	Rudder blade core, upper and lower halves	2	PETG, 15% gyroid
P14	Keel bulb casting pattern, forward and aft halves	1	PLA or PETG, 3 walls
P15	Electronics tray	1	PETG
P16	Battery strap anchor	4	PETG
P17	Penetrator boss pad	4	ASA or PETG, 100% infill
P18	Wing counterweight pod	1	PETG or ASA, 4 walls, 40% infill
P19	Backup tracker (YB3) deck cradle	1	ASA, 5 walls, 40% infill
P20	Navigation light clamp	1	ASA
P21	Limp-home controller case (potting shell)	1	PETG, 4 walls, 100% infill on the back
P22	Servo switchover pod (potting shell)	1	PETG
P23	RockBLOCK Plus bowl (hatch lid)	1	ASA, 5 walls, 40% infill
P24	Bilge pump spacer (mast tube)	1	PETG, 5 walls, 40% infill
P25	Tail detent magnet carrier	1	PETG, 100% infill
P26	Rudder servo arm for the 6 mm D-shaft	2	PETG, 100% infill

General notes

1. All dimensions in millimetres. Model origin: x aft from the bow, z up from the hull bottom at midships, y to port.
2. Design waterline (DWL) z = 106 at about 28.5 kg all-up mass (Rev B: z 90 at about 21 kg).
3. Printed parts: 0.4 mm nozzle, 0.2 mm layers, 4+ walls. Use ASA outdoors in UV; PETG is fine below deck.
   STL files are in print orientation; no supports needed except where a sheet says so.
4. Bought-in parts follow the Recommended Parts list in the build plan. Servo tab spacing, AS5600 board size and
   patch-antenna footprint vary by supplier: printed parts use slots and recesses rather than fixed hole patterns.
5. Full-size cutting templates (hull frames, foils, wing, hatch) are in Templates.pdf (tiled on Letter, 1:1)
   and as DXF files. Check the 100 mm scale bar on every printed page before cutting.
6. All fasteners and shafts 316 stainless. Bed every hull penetration in polyurethane sealant.
7. This is a hobby design, not a classed or certified vessel. Tank-test self-righting before going to sea.

Rev B (2026-11-08): wing counterweight moved from the wing's nose pocket to pod P18 on a 300 mm forward arm.
Rev C (2026-11-10), after the first real-weather Monte Carlo (leaks, wing damage and steering failures were the top losses):
  1. Hull foam-filled behind two glassed-in bulkheads (x 300, 1260), battery in its own sealed box: a leak can't swamp it (D03, D09).
  2. Twin rudders, 100 mm either side of the centreline, each with its own trunk, servo and tiller: either one steers alone (D05).
  3. Wing spar 30 × 26 carbon on 6806-2RS bearings (was 25 mm / 6805): about 1.8x the bending strength (D06, D07, P07, P08).
  4. Bulb D72 (was D68), about 9.2 kg, to hold stability at the higher mass (D04, P14).
  5. YB3 backup tracker platform P19 and all-round white light P20 on the stub; label areas (D01, D07, D09).
  Removed: thruster provision and the backup D-cell. Mass about 25 kg, DWL z 98, still self-righting from 180 deg.
Rev C1 (2026-11-11): electronics box raised onto a shelf (bottom z 118, was 60), above the 3 L sump at the low point of the dry bay;
  cable glands on the box top with drip loops, gel-filled connectors, desiccant. A leak no longer sits against the box (D03, D09).
Rev C2 (2026-11-11): keel fin rebuilt after a strength check: hardwood core, 6 × 6 oz glass and a carbon strip each side, 60 mm tenon into
  a keel-box socket, two ply floors across the hull (D04, T12, T13, T19, T20). About +1.1 kg. Proof-load before launch.
Rev C3 (2026-11-11), after the whole-boat structural review (docs/Structural_Review.md): 1. YB3 moved from the stern post to a well in the foredeck;
  post is now 20 × 2 fibreglass tube (D01, D07, P19). 2. Counterweight arm is a 12 mm solid fibreglass rod (D06). 3. Spring servo-saver and tiller
  stops on each rudder (D05). 4. Spar 30 × 22 carbon, bearings 130 apart in a longer P07 (D06, D07). 5. Second inner ply over the sump (D03). 26.9 kg, DWL z 102.
Rev C4 (2026-11-11), after the research review (docs/Research_Review.md): mast bearings are acetal (Delrin-type) with glass balls, floating on O-rings, with
  plain catcher lands in P07 for wave loads, and a V-ring seal on the spar (D07, P07). Notes added: fin glue-up and rod bedding (D04), rudder trunks (D05).
  ALL printed parts that see sun or sea: 2 coats of epoxy, then paint. Antifouling: 3 coats of ablative copper paint on hull, fin, bulb and rudders.
Rev C5 (2026-11-11): wing 800 × 375 (was 1000 × 300), same area, spar 995 long, after docs/Wing_Shape_Study.md (D01, D06, T15).
Rev D (2026-11-12; drafted as C6, a new letter because it adds a system: docs/Revisions.md), after docs/Electronics_Review.md:
  limp-home controller P21 (own GPS, compass and fused feed) takes the starboard rudder and the tail through switchover pod P22 if the main box
  dies; RockBLOCK Plus in bowl P23 in the hatch lid; main GPS to P05; every board coated; sensors, power boards and joints potted
  (D01, D03, D07, D08; docs/Limp_Home_Spec.md, docs/Potting_Spec.md). 27.8 kg, DWL z 104.
Rev E (2026-11-13; a new letter: it adds a system and a sensor), after docs/Reliability_Review.md: bilge pump P24 with keep-sailing leak response,
  membrane vents, Schrader test valve, humidity sensor; sealed rudder servos (SER-2000) in cradles P12 with arms P26 and spring pushrod links; tail servo potted,
  detent P25; weed fairing fin-to-bulb (D03, D04, D05, D06, D08, T21; docs/Design_RevE.md). 28.5 kg, DWL z 106.

SOLAR-SAIL OCEAN DRONE 1.5 m

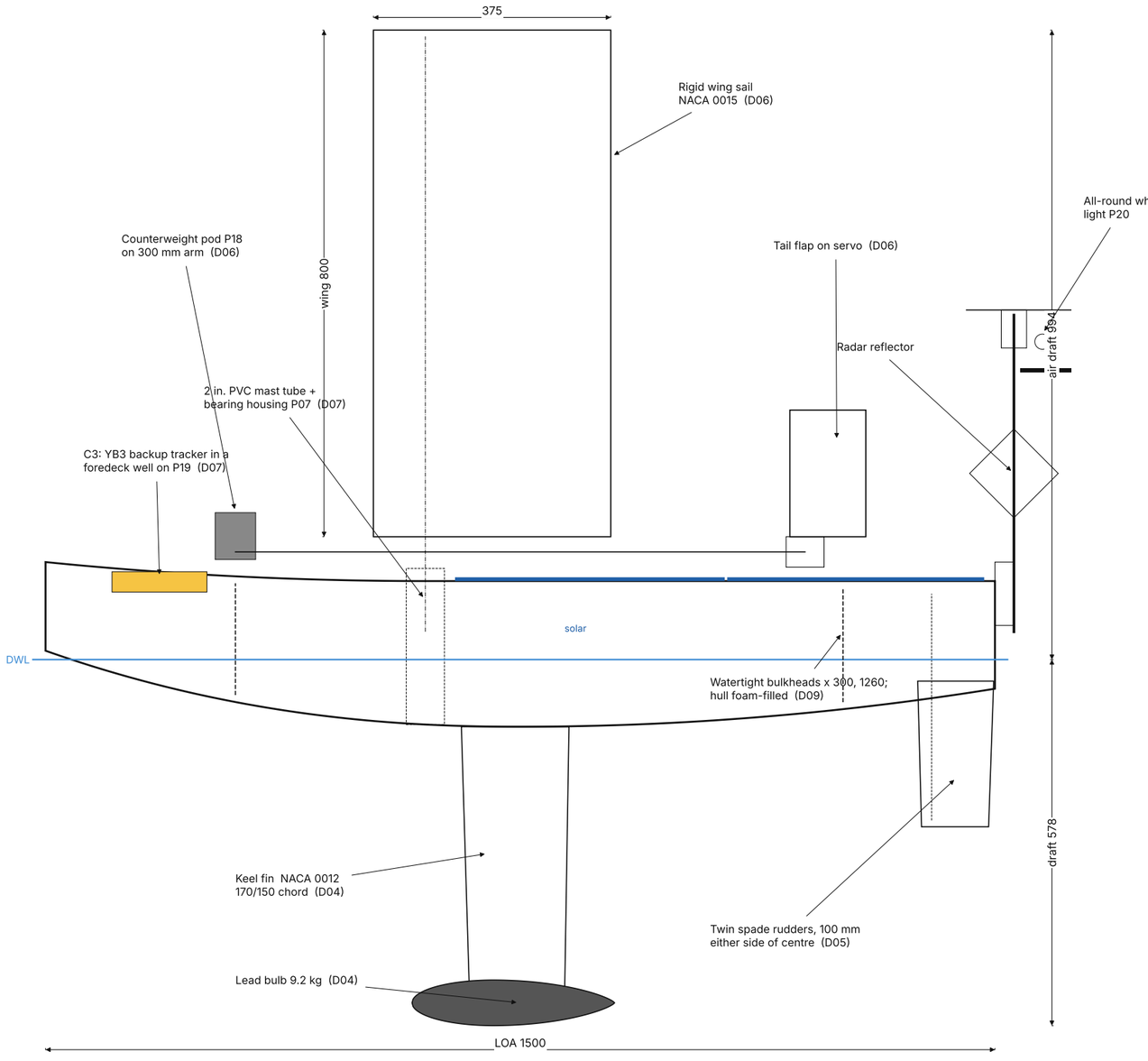
COVER, INDEX AND GENERAL NOTES

Units: mm

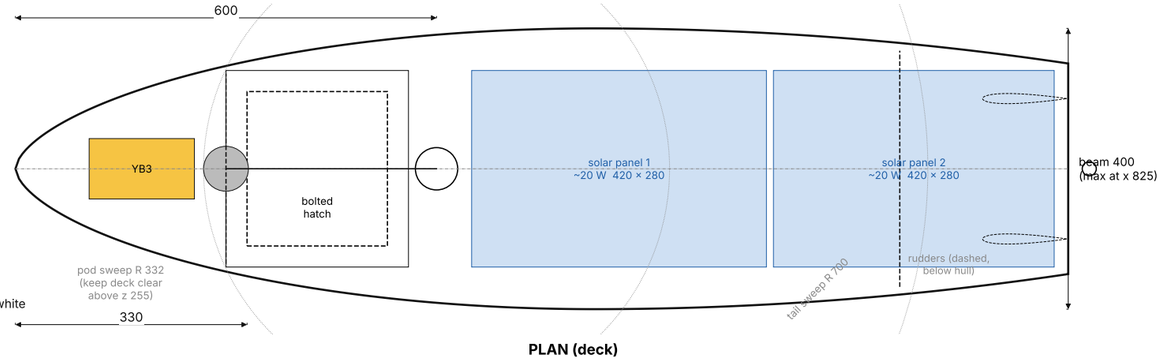
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET **D00**

REV **E**



PROFILE (starboard side, wing feathered aft)



PLAN (deck)

Key data (Rev C)

Length overall 1500 Beam 400 Hull depth 230
Draft 578 (DWL to bulb bottom) Air draft 994
Displacement about 28 kg Ballast 9.2 kg lead + about 0.8 kg rig counterweight
Wing 800 × 375 (0.30 m2; Rev C5), free-rotating on a 30 mm spar
Two rudders, two servos; hull foam-filled, two bulkheads
Solar about 40 W on deck Battery 192 Wh LiFePO4 in a sealed box
Self-righting from 180 deg (simulator): verify in a tank test (Gate 3)

SOLAR-SAIL OCEAN DRONE 1.5 m

GENERAL ARRANGEMENT

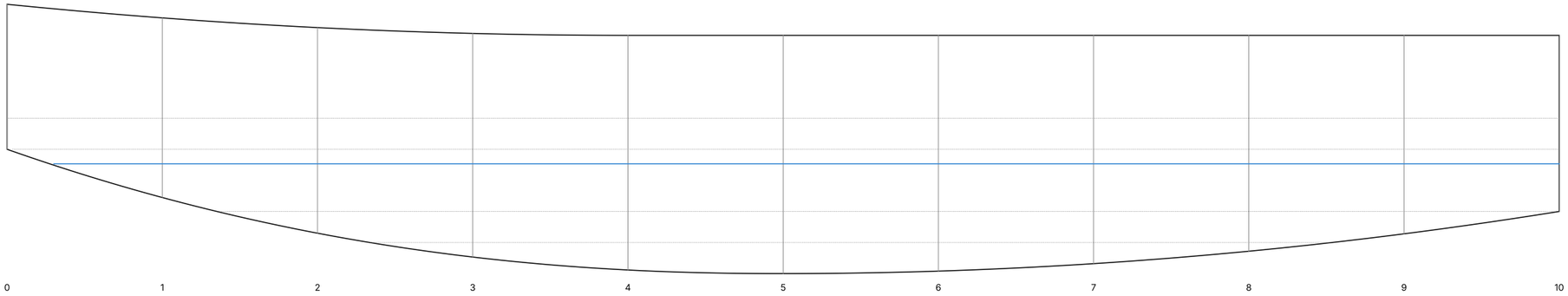
Units: mm Scale 1:8.5 on tabloid

Date 2026-11-13 Drawn with Claude Check all dims before cutting

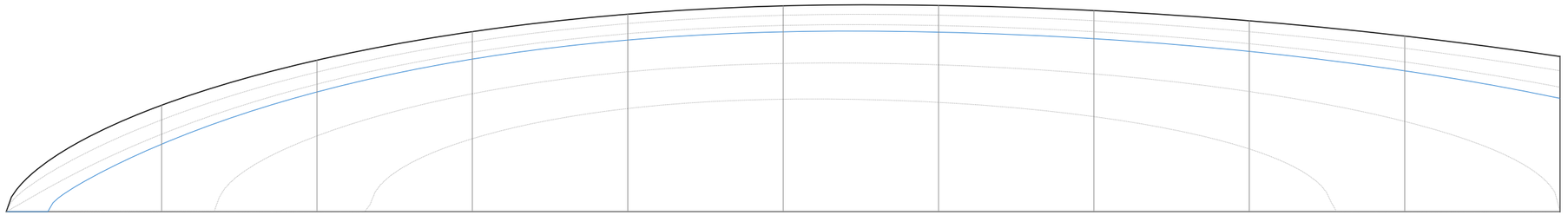
SHEET **D01**

REV **E**

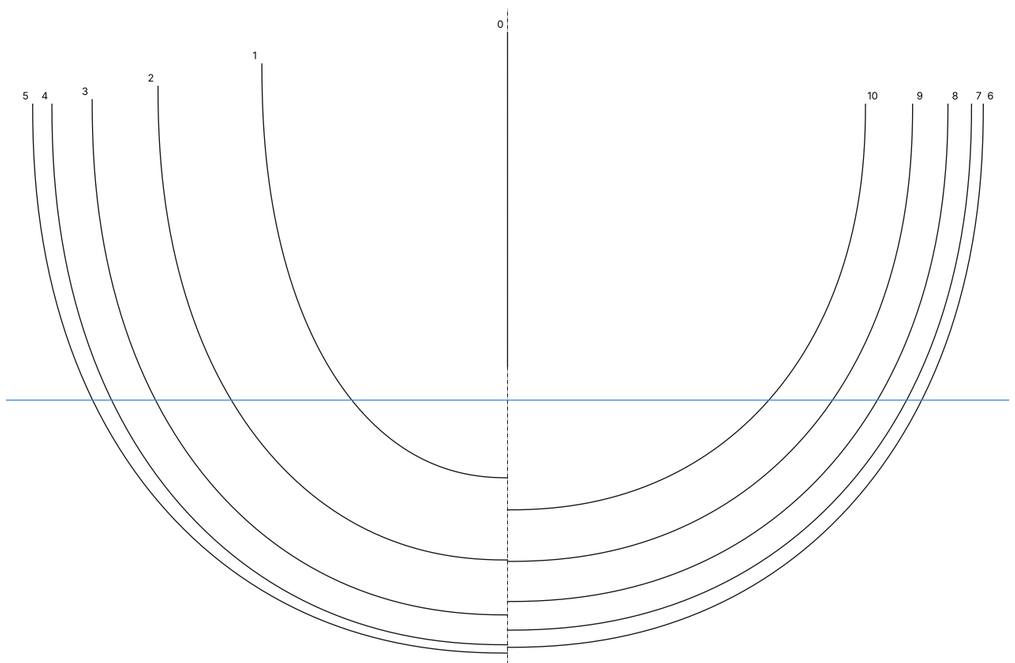
PROFILE



HALF-BREADTH PLAN (deck edge solid, DWL blue, waterlines z 30/60/120/150 dotted)



BODY PLAN (stations 0-5 left, 5-10 right)



Stn	x	keel z	deck z	1/2 b deck	1/2 b z30	1/2 b z60	1/2 b DWL	1/2 b z150
0	0	120.0	260.0	0.0	0.0	0.0	0.0	0.0
1	150	73.4	246.9	102.9	0.0	0.0	65.3	88.8
2	300	39.0	237.5	146.5	0.0	73.3	115.8	134.9
3	450	16.0	231.9	174.1	70.6	114.0	147.5	164.4
4	600	3.5	230.0	190.9	99.7	135.2	165.9	181.9
5	750	0.0	230.0	199.0	108.8	143.3	173.9	189.9
6	900	2.4	230.0	199.4	105.7	142.0	173.6	190.0
7	1050	9.6	230.0	194.4	91.9	133.2	167.2	184.6
8	1200	21.6	230.0	184.6	60.7	116.2	155.0	174.0
9	1350	38.4	230.0	169.8	0.0	87.3	136.2	158.0
10	1500	60.0	230.0	150.0	0.0	0.0	109.5	136.3

Section formula (exact; use it for intermediate points)

Half-breadth at height z: y = b(x) [1 - (1 - t)^2.2]^(1/2.2), t = (z - keel z) / (deck z - keel z)
b(x) = 200 [1 - (1 - x/825)^2]^0.6 forward of x 825; 200 - 50 ((x - 825)/675)^2 aft
Keel z = 120 (1 - x/750)^2.2 forward of x 750; 60 ((x - 750)/750)^2 aft
Deck z = 230 + 30 (1 - x/600)^2 forward of x 600; 230 aft

Construction: 10 mm H80 foam strips over 11 MDF frames (Templates T01-T11, cut on the dashed frame line = hull minus 11 mm), 2 × 6 oz glass outside, 1 × 6 oz inside, peel ply.
Rev C: hull form unchanged; DWL now z 106 at about 28 kg (the 1/2 b DWL column is at z 106).

SOLAR-SAIL OCEAN DRONE 1.5 m

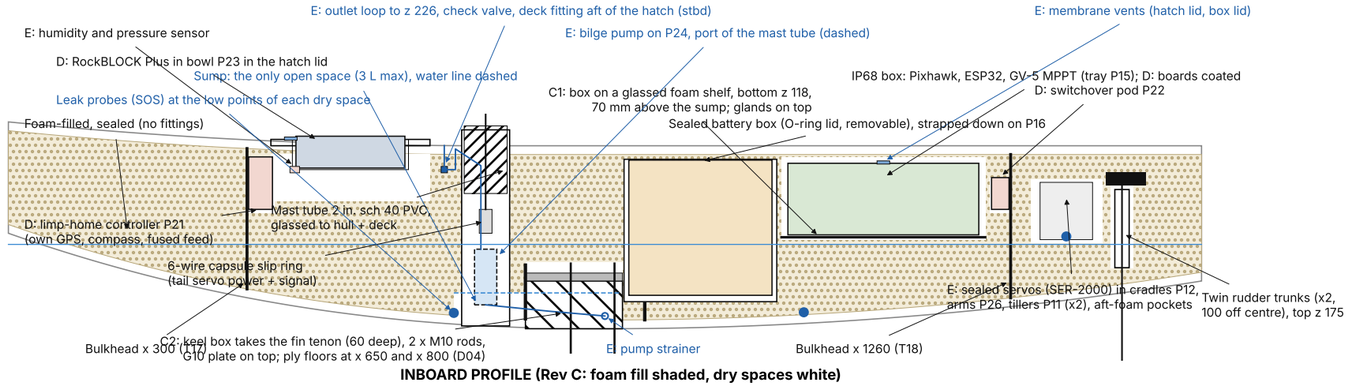
HULL LINES AND TABLE OF OFFSETS

Units: mm Profile & plan 1:5, body plan 1:2.5

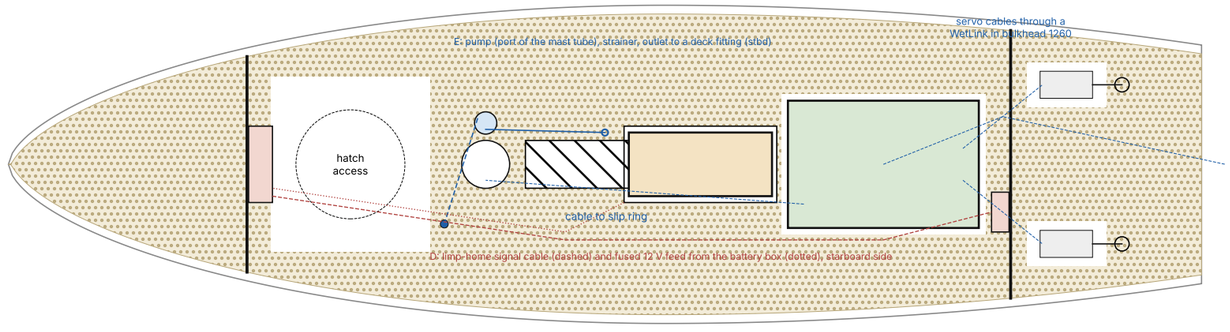
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET **D02**

REV **E**



INBOARD PROFILE (Rev C: foam fill shaded, dry spaces white)



INTERIOR PLAN (below deck, at z 120)

Systems notes (Rev C to E)

Inside volume: forward 6.6 L (solid foam), dry bay 55.4 L, aft 10.1 L (foam except the rudder trunks and servo pockets). XPS blocks at about 28 kg/m3, about 1.6 kg in all.

Fitted, removable blocks fill the dry bay around the boxes: aim for no more than 3 L of open space, so a leak can't add more than about 3 kg of water (12% of displacement). See D09.

Masses (approx, kg): hull + deck 4.5, bulkheads 0.2, foam 1.6, fin 2.1, bulb 9.2, rudders 1.0, servos + trunks 0.85 (E: sealed), rig 2.7, battery + box 2.35, electronics 1.2, Rev D limp-home + RockBLOCK Plus + wiring 0.85, Rev E pump, hoses, vents, sensor 0.2, solar 1.0, post items 0.95, YB3 0.45, keel box + floors 0.75, weed fairing 0.1, misc 0.6 → about 28.5 kg.

C3: the hull bottom over the sump (x 560-770) has no foam behind it: give it a second ply of 6 oz inside.

D: trim the fitted foam blocks round P21, P22 and the P23 bowl; the bowl leaves with the lid. Limp-home wiring and logic: docs/Limp_Home_Spec.md.

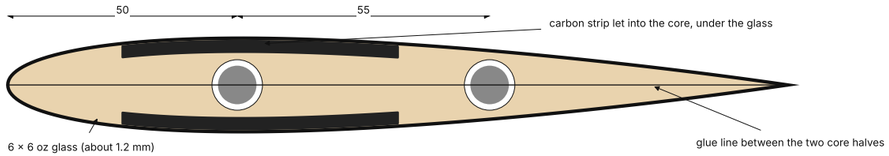
E: pump runs from the leak probe: keep sailing and pump; SAFE only if water passes 1.5 kg with the pump running. Cut the fitted foam round the pump and hoses. Check valve arrow toward the deck.

E: enlarge each aft-foam servo pocket to 100 × 56 for the SER-2000 enclosure; the servo cable comes through its own penetrator, so no water reaches the dry bay if the housing floods.

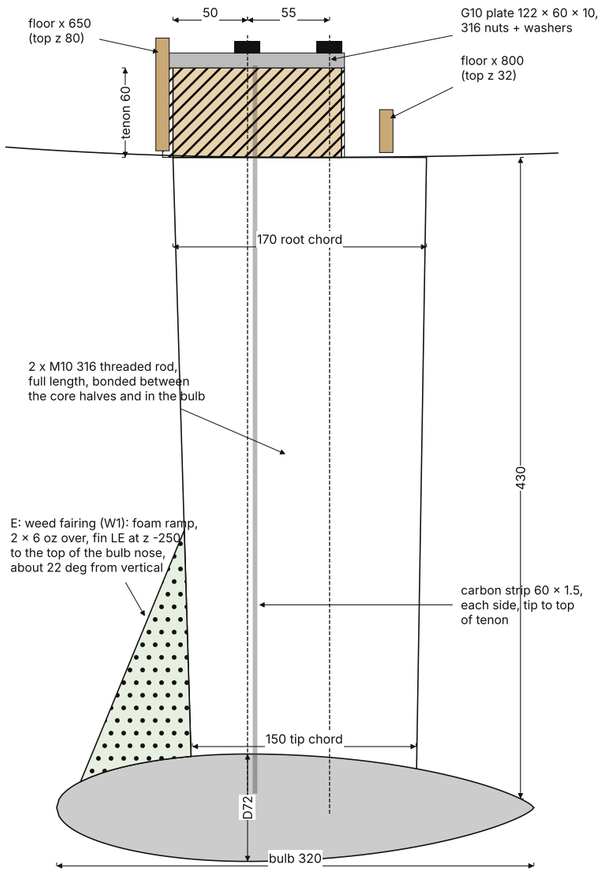
SOLAR-SAIL OCEAN DRONE 1.5 m

INTERIOR ARRANGEMENT AND SYSTEMS

Units: mm Scale 1:7.5 Date 2026-11-13 Drawn with Claude Check all dims before cutting	SHEET **D03**	REV **E**



FIN ROOT SECTION NACA 0012 (1:1.3)



FIN + BULB ELEVATION

Fin construction (Rev C2)

1. Core: two planks of straight-grained hardwood (white oak, ash or sapele), each 10.5 thick x 175 × 495, grain along the span.
2. Rout two half-round grooves (R5.5) in each plank at 50 and 105 from the root LE, parallel to the LE. Glue the planks together round the 2 x M10 316 rods with thickened epoxy. Rods project 30 above the tenon and 60 below the tip.
3. Plane to templates T12 (root) and T13 (tip). The top 60 mm is the tenon: keep the root section, cut it back to 113 long (x 657-770).
4. Rout a recess 60 wide x 1.5 deep each side, 25-85 from the LE, full length incl. the tenon. Bond in unidirectional carbon (pultruded strip, or 5 layers of 300 g tape).
5. Glass 6 × 6 oz each side, layers 2 and 5 at +/-45 deg, wrapped round the LE; fair, then 2-part epoxy primer.

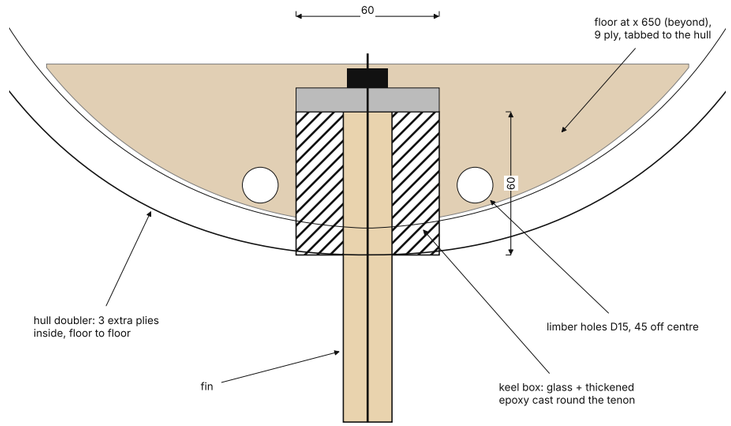
Keel box, floors and hull

6. Floors: 9 mm marine ply to T19 (x 650) and T20 (x 800), 2 × 6 oz each side, tabbed to the hull with 3 × 6 oz, 50 mm each way.
7. Doubler: 3 extra plies of 6 oz inside the hull bottom between the floors, out to 120 either side.
8. Keel box: wax the tenon, stand the fin in the hull slot, build 6 plies of glass round the tenon and fill out to 60 wide with thickened epoxy between the floor at x 650 and a ply end plate at x 772.
9. G10 plate 122 × 60 × 10 on top, 316 nuts + washers, 15 N.m, thread-lock. Seal the hull slot with polyurethane.
10. PROOF TEST before paint: Field Test K (60 kg on the bulb).

C4: sand glue faces to 80 grit and wipe twice with alcohol (oak especially); glue the core with G/flex. Bed the rods fully in epoxy, no voids; seal the end grain at tip and tenon.

E: after the bulb is on, fill the step between the fin LE and the bulb nose with the fairing: XPS shaped to T21, bonded, 2 × 6 oz lapped 30 mm onto fin and bulb. Weed sliding down the fin runs off the nose instead of hooking in the step (about 0.1 kg).

from nose	diameter
0	0.0
20	38.6
40	52.3
60	61.0
80	66.7
100	70.3
120	71.9
140	71.8
160	70.5
180	68.0
200	64.3
220	59.2
240	52.7
260	44.6
280	34.4
300	21.3
320	0.0



SECTION AT x 735, LOOKING AFT (1:2.5)

Why (docs/Keel_Structure_Loads.md)

The foam-cored fin with 3 × 6 oz was good for about 113 N.m at the root: 3 times the load of the bulb with the boat on its side (36 N.m). A breaking-wave strike is estimated at 65 N.m typically and 350 N.m once in 1,000. This fin is estimated at 1,100 N.m and the tenon joint at about 700 N.m. Both are hand calculations: the proof test decides.

Bulb (Rev C: D72, was D68)
Volume 808 cm3 → 9.2 kg of lead. Make a plaster mould from pattern P14a/b (printed), cast outdoors with PPE, then drill 2 × 12 mm holes 60 deep at the rod positions (91 + 37 and 91 + 92 mm from the nose, measured on the slot) and bond the rods in with epoxy. Fair with epoxy filler; seal with 2 coats epoxy barrier coat.

SOLAR-SAIL OCEAN DRONE 1.5 m

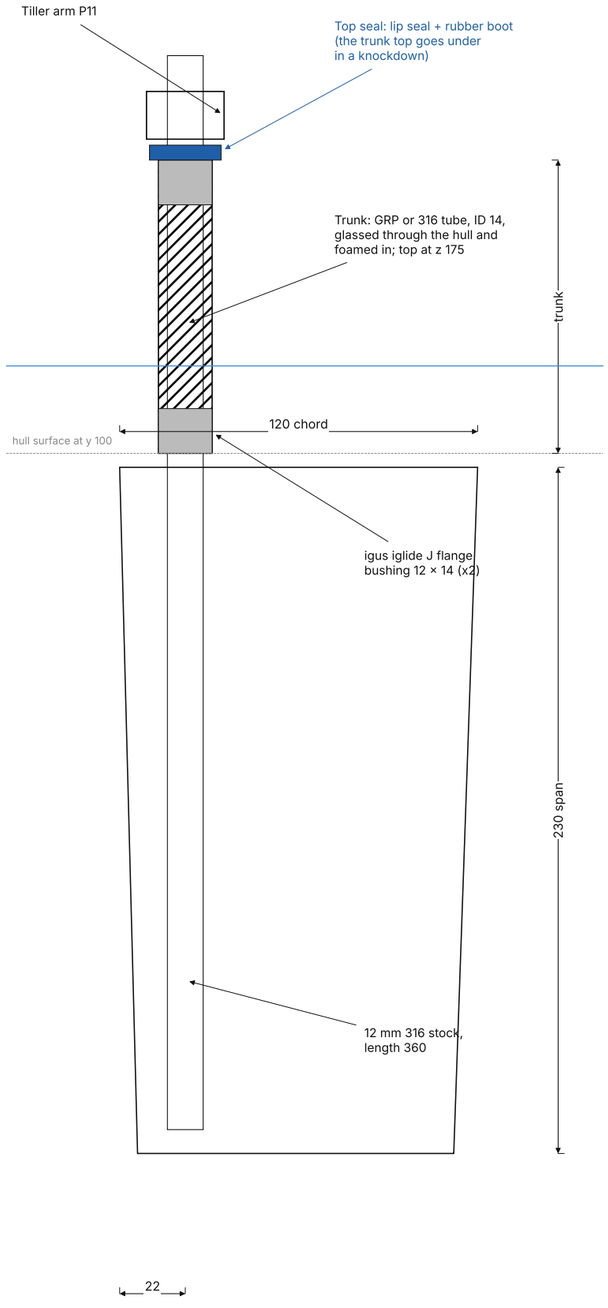
KEEL FIN, KEEL BOX, LEAD BULB AND WEED FAIRING

Units: mm Scale 1:4, sections as noted

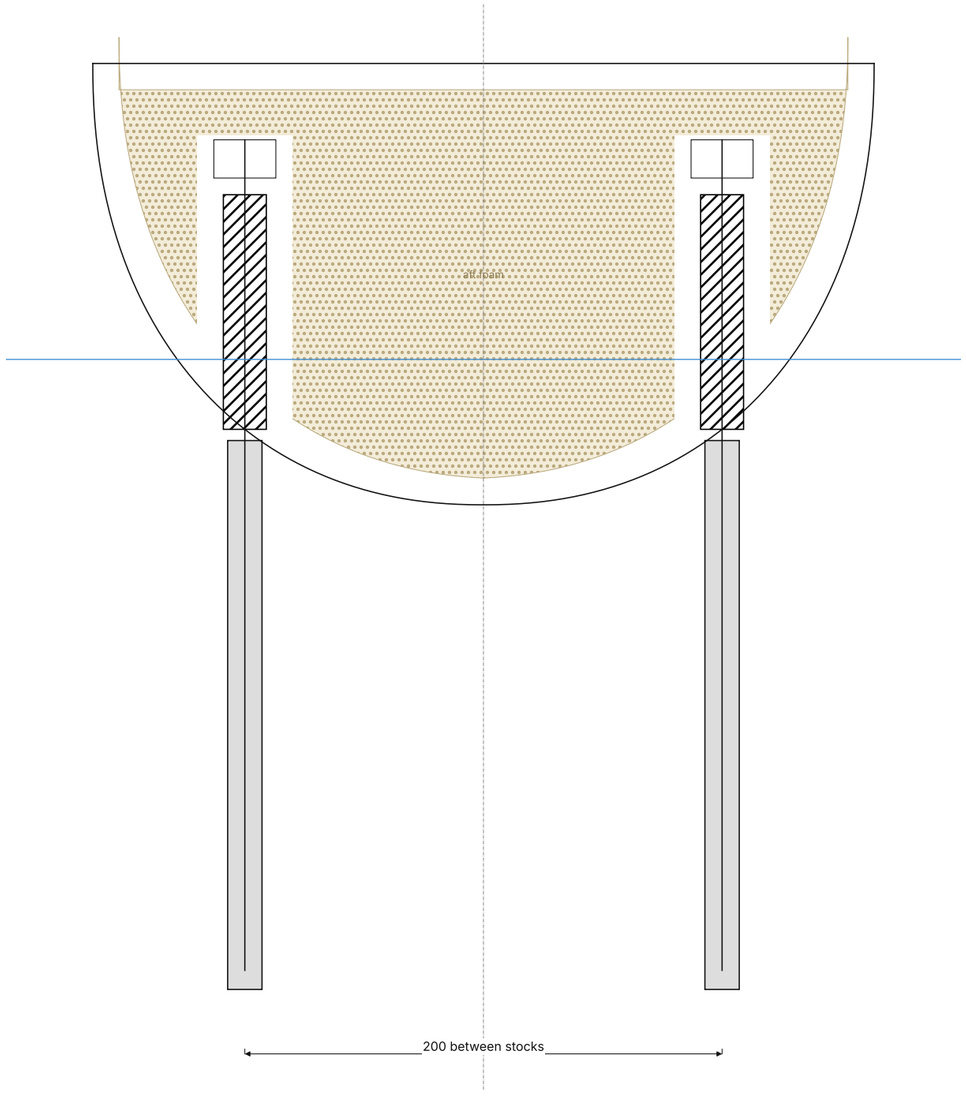
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D04

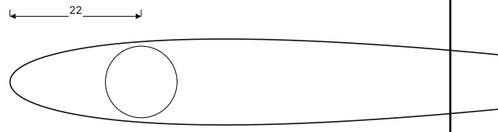
REV E



ONE RUDDER, ELEVATION (looking to port; both are the same)



SECTION AT x 1400, LOOKING FORWARD (1:2.5)



BLADE SECTION NACA 0012 (1:1)

Build (make two of everything)

1. Print cores P13a/b (two sets), slide on each stock, pin with 4 mm 316 pins, epoxy the joints.
2. Glass 2 × 6 oz over each blade, wrapping the tip; fair to template T14.
3. Balanced: about 15% of the area is ahead of the stock, so each servo needs little holding torque.
4. Stocks: 12 mm 316 round bar, 360 long; 3 mm cross-hole for each tiller pin.
5. Trunk bottoms flush with the hull; glass a 40 mm fillet pad where each meets the curved bottom.
6. Each rudder has its own servo (P12), tiller (P11) and cable. Either one alone can steer the boat.
7. ArduPilot: two outputs set to GroundSteering.
8. C3/E: a breaking wave can force a blade over with 4 x the servo's stall torque. E: the sealed servo (SER-2000, 6 mm D-shaft) takes arm P26; put a spring overload link in each pushrod (316 spring pack, gives at about 2 x working load) in place of the 25T servo-saver. Keep the G10 stops at +/-40.
9. C4: 316 corrodes in trapped, still sea water. Pack each trunk with waterproof grease; pull and inspect the stocks for pitting after the shakedown test. If one servo jams, the mission manager trims the other to compensate; if one goes limp it trails.
10. Stop each rudder at +/-35 deg with the servo end points. Check both blades clear the bulb's wake.

E: steer on heading error and yaw rate low-passed at about 4 s (tune in Test I); servos stay powered.

E: raked rudders (W3) NOT adopted: a 35 deg rake puts the tip 160 mm aft, past the transom and off the stock.

SOLAR-SAIL OCEAN DRONE 1.5 m

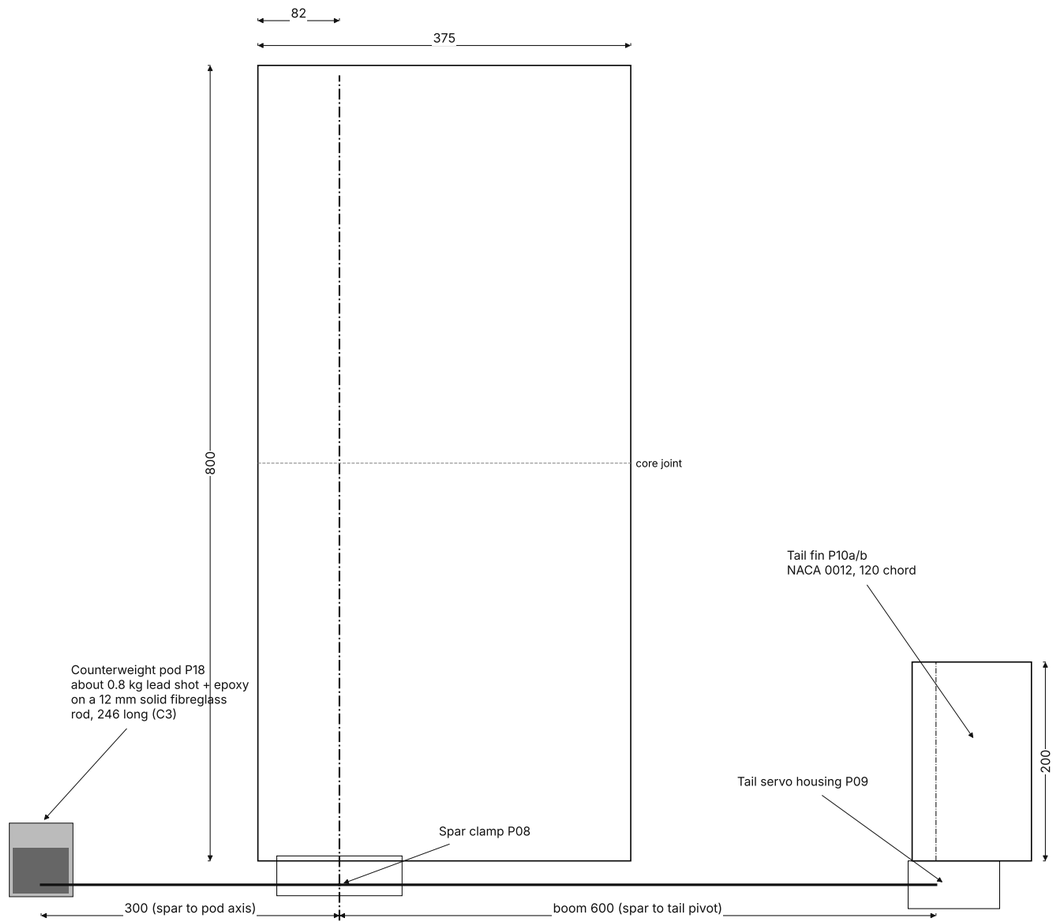
TWIN RUDDERS, STOCKS AND STEERING (REV E: SEALED SERVOS)

Units: mm Scale 1:2

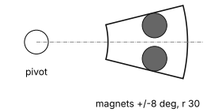
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET **D05**

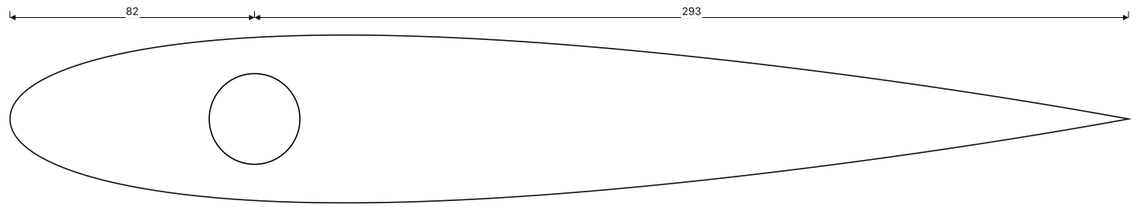
REV **E**



WING, TAIL + COUNTERWEIGHT ELEVATION



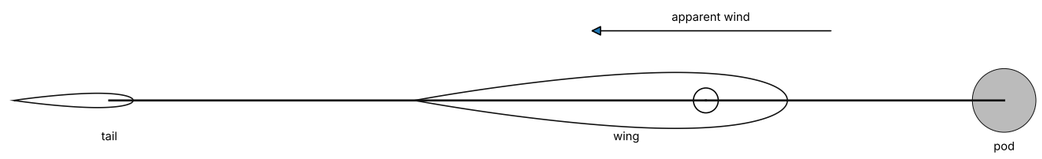
E: TAIL DETENT P25, PLAN (1:1.5)



WING SECTION NACA 0015 (1:2)

Wing build

Core: two 400 mm XPS blocks hot-wire cut with template T15 (same root and tip), bored 30.4 mm for the spar (Rev C).
Skin: 2 × 6 oz glass, vacuum bagged if possible. Root and tip: 3 mm ply caps. Spar: 30 × 22 carbon tube, 995 long.
Rev C5: wing 800 × 375 (was 1000 × 300), same 0.30 m2, spar 82 from the leading edge: about half the load when a knockdown slaps the wing on the water, 4% slower.
Balance: assemble wing, tail, boom, arm and pod; add lead shot + epoxy to P18 until the rig sits level on the spar at any heading.
Expect about 0.85 kg (C5). The pod sweeps a 332 mm radius at z 264-338, about 15 mm over the hatch bolts: keep anything on deck in that circle below z 255.
Tail angle sets the wing's angle of attack: about +/-8 deg tail gives the working range; 0 deg = wing feathered.
E: tail servo is a DFRobot magnetic-encoder servo potted in P09. Detent P25 on P09 (two 6 × 3 magnets at r 30, +/-8 deg) and a steel disc
in the underside of P10: if the servo lets go, the tail snaps to +/-8 deg and the wing keeps driving instead of feathering (the one certain-loss failure).



PLAN OF ROTATING RIG (1:6)

SOLAR-SAIL OCEAN DRONE 1.5 m

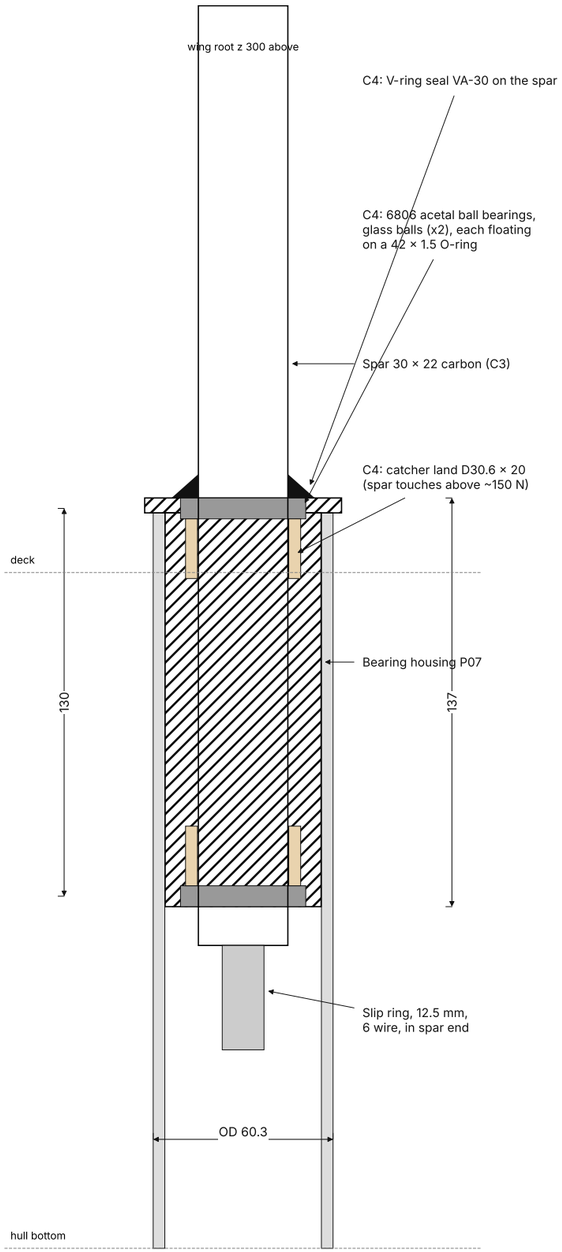
WING SAIL AND TAIL

Units: mm Elevation 1:6, plan 1:6, section 1:2

Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET **D06**

REV **E**



MAST STEP SECTION (1:2)

STERN STUB (1:5)

Notes

Spar: 30 × 22 carbon tube, 995 long (z 105 to 1100). Rev C3: 1.6 x the strength of 30 × 26. Bearings 130 apart in the longer housing P07; give the mast tube a 60 mm wide glass collar where it meets the deck.
C4 bearings: acetal rings and glass balls cannot rust or seize (stainless ones do, in still salt water) but carry only about 170 N. Each floats 0.3 in its seat on an O-ring; in a wave strike (several kN) the spar bears on the two plain lands. Grease the lands. A Delrin thrust collar clamped on the spar rests on the upper bearing and carries the rig.
No drain in the mast tube: the slip ring sits at its foot.
Slip ring in the spar's lower end carries 3 wires to the tail servo; it must turn freely through 360 deg.
Lock the spar against lifting with a 316 collar under the lower bearing.

Post (C3): 20 mm OD x 2 wall pultruded fibreglass tube, 500 long (not aluminium or carbon: it must bend, not kink, when a wave jolts the boat); cables run inside it to the vane cap and down through a WetLink in the transom.
Vane shaft: 8 mm 316, 75 long, 2 x S608-2RS bearings.

On the post, bottom to top: radar reflector (z 330-470), main GPS on P05 (z 560; Rev D, was the Iridium patch), light P20 (z 600), vane (z 650).

YB3 (C3): in a glassed well in the foredeck, 176 × 106 × 38 deep, centred at x 180, cut into the solid bow foam, with a 10 mm scupper slot to each deck edge. Cradle P19 on the well floor; the YB3's top is flush with the deck, 25 mm under the tail boom. It has its own battery and antenna: no cables.

SOLAR-SAIL OCEAN DRONE 1.5 m

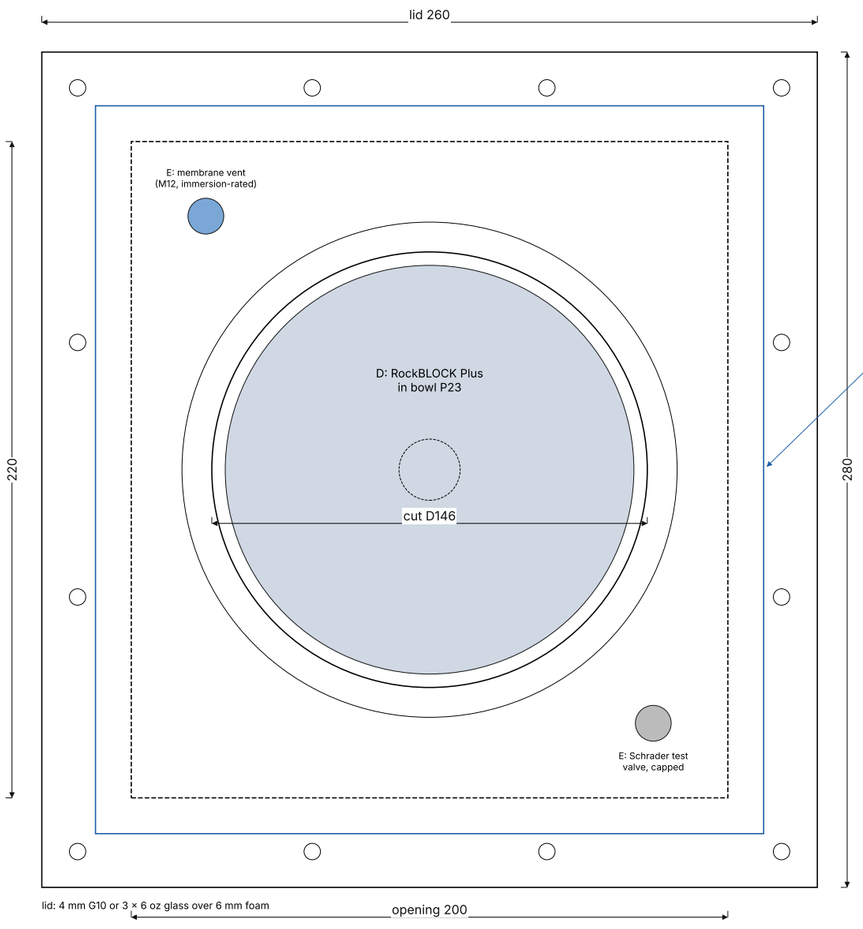
MAST STEP, SPAR AND STERN STUB

Units: mm Scale 1:2 and 1:5

Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET **D07**

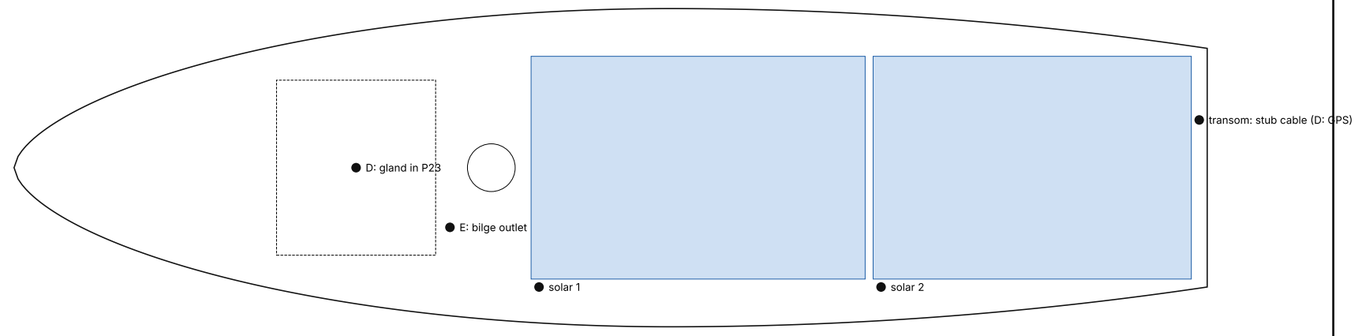
REV **E**



HATCH LID AND OPENING (1:2)

Hatch

12 x M5 316 bolts into bonded-in M5 inserts (or nuts glassed on a doubler).
Doubler under the deck: 30 mm wide frame of 6 mm foam + 2 × 6 oz glass.
Torque evenly; grease the O-ring with silicone grease; tape the seam for the crossing.
Template T16 gives the opening, bolt holes and lid outline at 1:1.
D: the RockBLOCK Plus sits in bowl P23 bonded into a D146 hole in the lid, its top 2 mm proud and below z 255 (counterweight pod sweep). Its cable passes an M20 IP68 gland in the bowl floor to an inline IP68 connector inside: unplug it before lifting the lid. Sealant fillet round the puck so no water stands over the antenna.
The Iridium patch and its cable on the stern post are deleted; the post cable now carries the main GPS (P05).
E: a membrane vent and a capped Schrader valve in the lid, both outside the bowl and the bolt line. Pressure-decay test before every launch: bicycle pump to 0.1 bar through the valve, no measurable drop in 30 min (Field Test G).
E: bilge outlet: 1/4 in. through-deck fitting at x 548, 75 to starboard, bedded; hose check valve just below it.



DECK PENETRATIONS (P17 pads + WetLink)

SOLAR-SAIL OCEAN DRONE 1.5 m

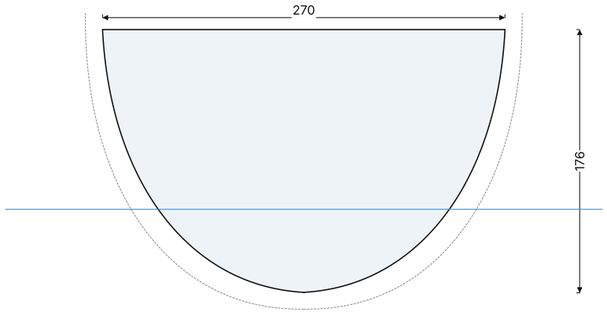
DECK HATCH, SOLAR LAYOUT AND PENETRATIONS

Units: mm Hatch 1:2, deck 1:7.5

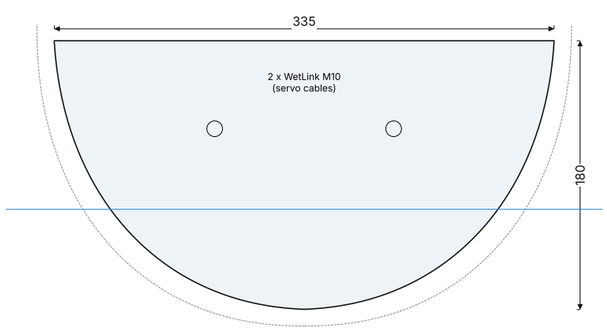
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET **D08**

REV **E**



BULKHEAD x 300 (T17, 1:4)



BULKHEAD x 1260 (T18, 1:4)

Bulkheads

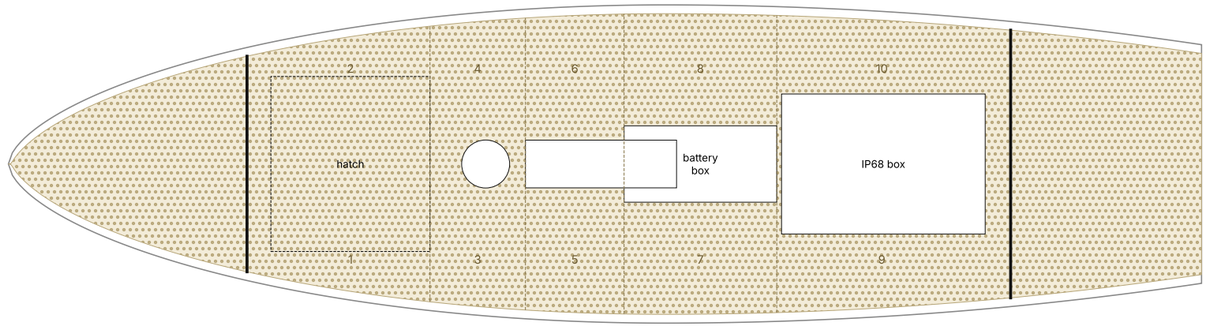
10 mm H80 foam, 2 × 6 oz glass each side. Cut on the solid line (hull inside, 11 mm in from the outside).
Fit after the hull is off the mould, before the deck goes on. Glass in with 50 mm tapes both sides,
on a thickened-epoxy fillet. They make three spaces: forward (solid foam), dry bay, aft (trunks and servos).
No hatches or cables through bulkhead x 300. Bulkhead x 1260 carries the two servo cables in WetLinks.

Leak test, each space: 0.1 bar of air through a temporary fitting with soapy water on every joint and
penetration, then a 72-hour submersion of the finished hull before Field Test F (Failure_Rates.md, section 5).

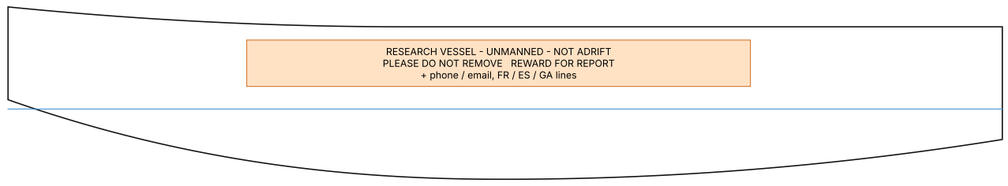
Space (x)	Inside L	Kept dry for	Foam L	Free L
Forward 0-300	6.6	nothing	6.6	0
Dry bay 300-1260	55.4	battery box 3.3, IP68 box 4.8, mast tube, keel box, cable runs 2.1	42.3	<= 3
Aft 1260-1500	10.1	2 trunks, 2 servo pockets about 1.5	8.6	<= 0.5

Foam fill

About 57 L of extruded polystyrene (XPS, 25-30 kg/m3, closed cell), about 1.6 kg.
Forward space: blocks bonded in solid with expanding PU adhesive before the deck goes on.
Aft space: blocks bonded around the trunks; leave fitted pockets for the two servos (lids of foam, taped).
Dry bay: numbered, removable blocks cut to fit around the boxes, taped together; they come out through the
hatch for maintenance. Fill until no more than 3 L of open space is left (pour water in to check, then dry).
Why: a leak can then add at most about 3 kg of water (12% of displacement) and can't swamp the boat.
C1: leave the open space as one sump at the low point (x 560-770, below z 45). Everything above it is foam, so the water
stays 70 mm below the electronics shelf upright and at 30 deg of heel. Test J in Field_Tests.md checks this.



DRY-BAY BLOCK LAYOUT (plan at z 120; numbers = removal order through the hatch)



LABEL AREAS (each side of the hull, both faces of the wing; 1:9)

Labels and colour

Fluorescent orange deck, wing panels and label areas. Text black, 25 mm capitals; vinyl, sealed under clear epoxy.
Wing: same text on both faces, lower half, so it reads with the boat heeled. Wording: Launch_Plan.md.

SOLAR-SAIL OCEAN DRONE 1.5 m

WATERTIGHT SUBDIVISION, FOAM FILL, BATTERY BOX, LABELS

Units: mm Bulkheads 1:4, plan 1:7.5, labels 1:9

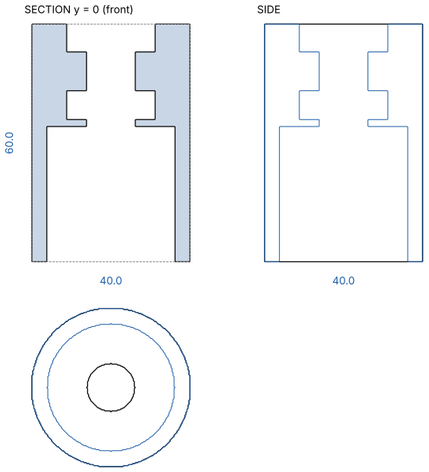
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D09

REV E

P01 Wind-vane sensor housing

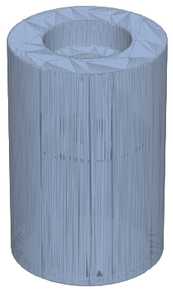
File: stl/P01_vane_housing.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
OD	40
Height	60
Bearing seats	2 x D22.1 × 7
Cavity	D32 × 34

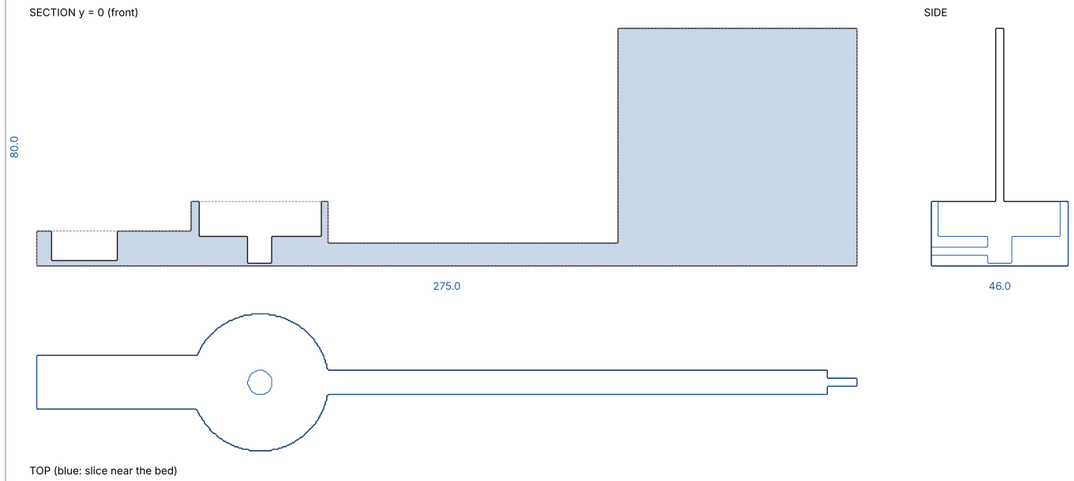
Material: ASA or PETG, 100% walls 4

- Two S608-2RS stainless bearings (8×22×7) press in from the top.
- AS5600 board sits on cap P03 inside the cavity; magnet on shaft end faces it.
- Bottom cavity is closed by cap P03, bonded with thickened epoxy.



P02 Wind-vane arm with fin and counterweight pocket

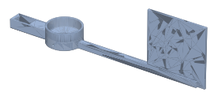
File: stl/P02_vane_arm.stl Views at 1:2 Print orientation: z up from the bed



Feature	Size
Arm length	190 + 75
Fin	80 × 80 × 3
Hub	D46 × 22
Shaft bore	D8.1 × 9 blind

Material: ASA or PETG

- Printed upside-down: z=0 face is the top of the hub in service.
- Skirt (D41 inside) overlaps the housing top as a rain labyrinth.
- Fill the pocket with lead shot + epoxy until the arm balances on the shaft.
- M3 set screw (tap the D2.5 radial hole) locks the hub on the 8 mm shaft.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

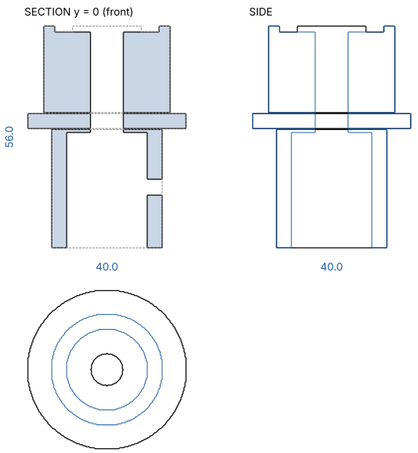
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D10

REV E

P03 Sensor cap and stub-tube socket

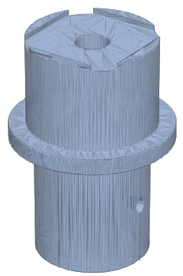
File: stl/P03_sensor_cap.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Spigot	D31.6 × 22
Flange	D40 × 4
Socket	OD28, ID20.4 × 30

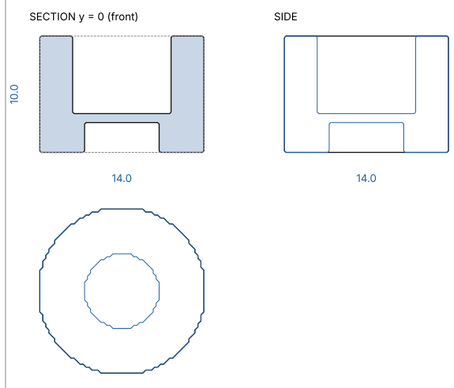
Material: ASA or PETG

- Spigot (D31.6) slides into the P01 cavity; bond and seal with epoxy.
- Recess on top takes the AS5600 board (glue down); cable runs down the D8 hole into the 20 mm stub tube.
- Socket fits the 20 mm stub tube; M4 set screw (D3.3 tap hole).
- Printed socket-down as modelled.



P04 Magnet holder for vane shaft

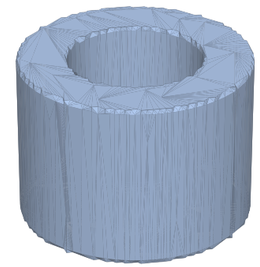
File: stl/P04_magnet_holder.stl Views at 1:0.5 Print orientation: z up from the bed



Feature	Size
OD	14
Height	10
Magnet pocket	D6.2 × 2.7
Shaft bore	D8.05 × 6.5

Material: PETG

- Press onto the bottom end of the 8 mm 316 shaft below the lower bearing.
- Glue a 6 × 2.5 mm diametric magnet (AS5600 type) into the bottom pocket.
- Also acts as the retaining collar that stops the shaft lifting out.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

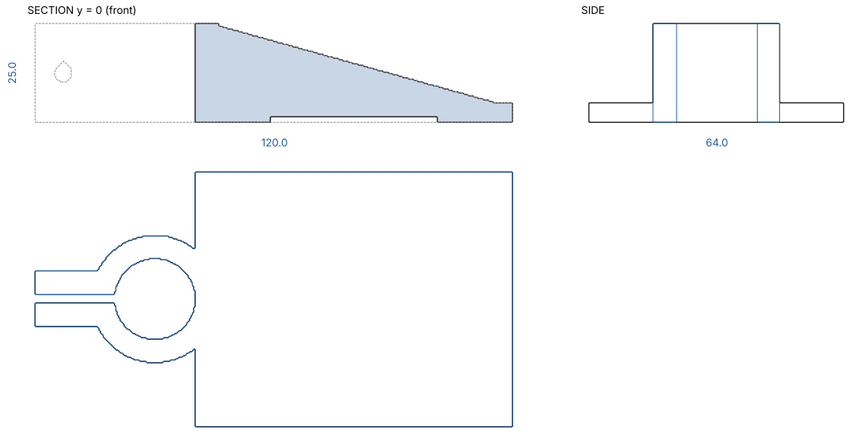
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D11

REV E

P05 Antenna platform clamp

File: stl/P05_antenna_platform.stl Views at 1:1.5 Print orientation: z up from the bed

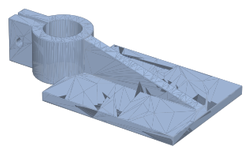


TOP (blue: slice near the bed)

Feature	Size
Platform	80 × 64 × 5
Collar	OD32, ID20.3 × 25

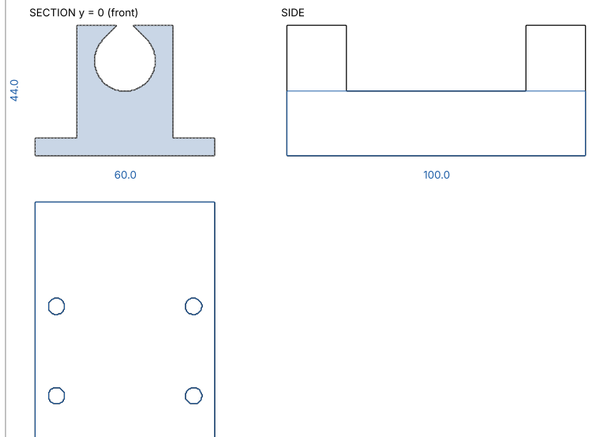
Material: ASA

- Clamps to the 20 mm stub tube at 560 mm above the hull bottom.
- Rev D: carries the main GPS + compass (Holybro Micro M10, potted in soft silicone, patch up), not the Iridium patch (the RockBLOCK Plus has
- Squeeze slot closed with an M4 bolt (D4.3 hole) through the ears.



P06 Transom bracket for 20 mm stub tube

File: stl/P06_transom_bracket.stl Views at 1:2 Print orientation: z up from the bed

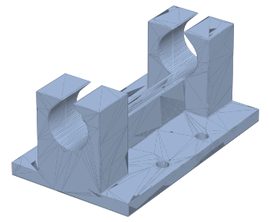


TOP (blue: slice near the bed)

Feature	Size
Plate	60 × 100 × 6
Tube offset from transom	32
Blocks	32 × 20 × 44 (x2)

Material: ASA, 6 walls

- Plate bolts to the transom with 4 x M5 316 bolts, bedded in sealant over a glass-reinforced pad.
- Tube bores are teardrop-shaped so they print without support.
- M4 set screws (D3.3 tap holes) lock the tube in both blocks.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

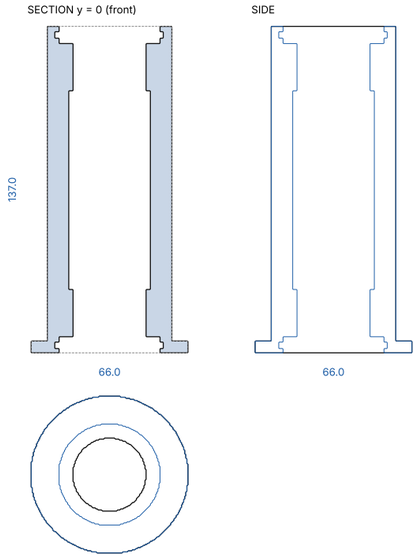
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D12

REV E

P07 Mast bearing housing with catcher lands

File: stl/P07_mast_bearing_housing.stl Views at 1:2.5 Print orientation: z up from the bed



Feature	Size
OD	52.3
Flange	D66 × 5
Length	137
Bearing seats	2 x D42.6 × 7, O-ring groove D45.4 × 2.6
Catcher lands	2 x D30.6 × 20
Clearance bore	D34

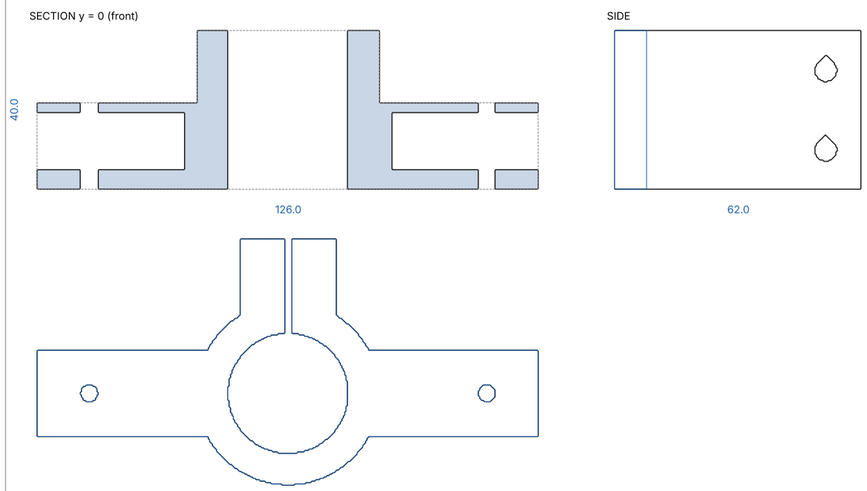
Material: PETG or ASA, 100% infill

- Bonds into the top of the 2 in. sch 40 PVC mast tube (ID 52.5); flange rests on the tube end. Printed flange-down as modelled.
- Rev C4: two 6806 acetal ball bearings with glass balls (30×42×7), 130 mm apart. They cannot rust or seize, but carry only about 170 N each.
- Each bearing floats 0.3 mm in its seat on a 42 × 1.5 O-ring (groove in the seat). Beside each is a plain land, D30.6 × 20:
- above about 150 N the spar moves 0.3 mm and bears on the lands, which take wave loads of several kN. Wax or grease the lands.
- If you can bore Delrin, open the lands to D36 and bond in Delrin sleeves 36 × 30.6 × 20 (lower friction than the print).



P08 Spar clamp with tail-boom and counterweight-arm sockets

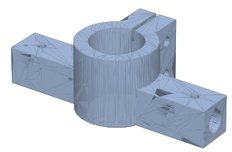
File: stl/P08_spar_clamp_two_sockets.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Collar	OD46, ID30.2 × 40
Boom socket (aft)	D12.2 × 37
Arm socket (fwd)	D12.2 × 37
Ears (side)	24 × 22
Overall	126 × 62 × 40

Material: ASA or PETG, 100% infill

- Rev C: clamps the 30 mm spar just below the wing root (2 x M5 bolts through the side ears). Rev B was 25 mm.
- Aft socket (+x): 12 mm carbon tail boom, 40 deep. Forward socket (-x): counterweight arm, 40 deep (Rev C3: 12 mm solid fibreglass rod).
- Pin each tube with an M4 bolt through the D4.3 vertical holes. Both tube axes are 11 mm above the bottom face.
- Line the forward socket up with the wing chord: the arm must point straight at the wing's leading edge.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

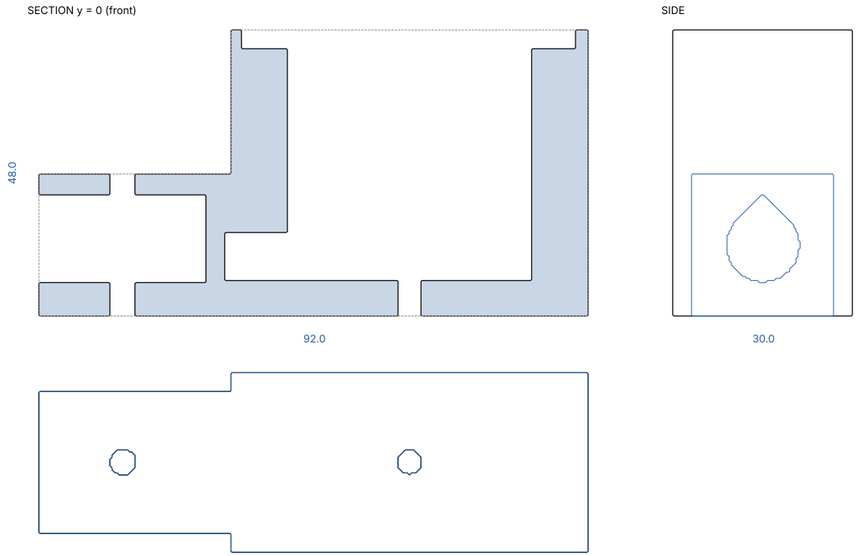
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D13

REV E

P09 Tail servo housing (boom end)

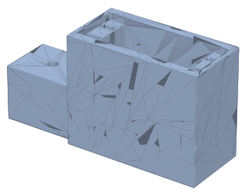
File: stl/P09_tail_servo_housing.stl Views at 1:1 Print orientation: z up from the bed



Feature	Size
Overall	92 × 30 × 48
Servo pocket	41.2 × 21.2 × 42
Boom socket	D12.2 × 28

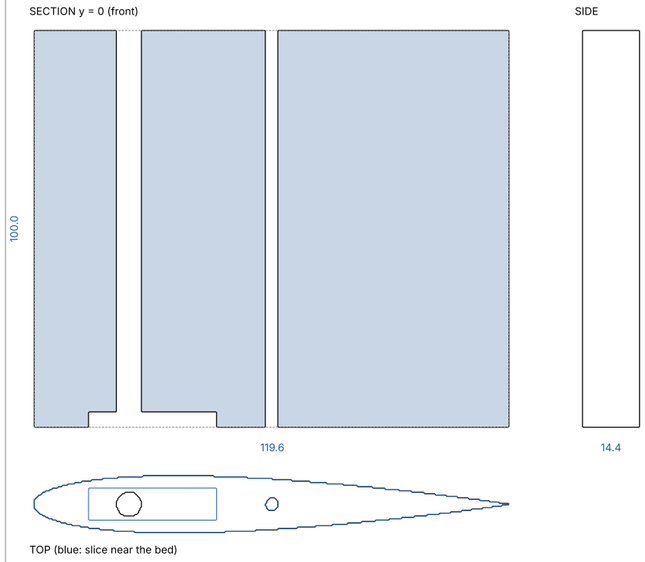
Material: ASA or PETG

- Slides on the aft end of the 12 mm boom; pin with M4 (D4.3 hole).
- Rev E: holds a DFRobot magnetic-encoder servo (25 kg) output-up, potted:
- urethane over the case seams and cable entry, silicone round the output.
- The tail fin P10 bonds to the servo horn; detent P25 bonds on top, aft.
- Tab screw slots allow 47.5-51.5 mm tab spacing: check your servo before printing.
- Pocket has a D4 drain hole; cable exits through the slot toward the boom.



P10 Tail fin, lower and upper halves [tail fin lower]

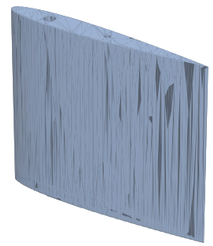
File: stl/P10a_tail_fin_lower.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Chord	120
Span	2 × 100
Thickness	14.4
Pivot	D6.2 at 24 from LE

Material: PETG or ASA, 0.2 mm layers

- NACA 0012, 120 mm chord, 200 mm span in two 100 mm halves printed standing up.
- D6.2 bore at 24 mm from the leading edge takes a 6 mm carbon pivot rod through both halves.
- D3.2 bore at 60 mm takes a 3 mm carbon rod to align and stiffen the joint.
- Lower half: 32 × 8 × 4 recess on the bottom face to bond an aluminium 25T servo horn.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

Date 2026-11-13 Drawn with Claude Check all dims before cutting

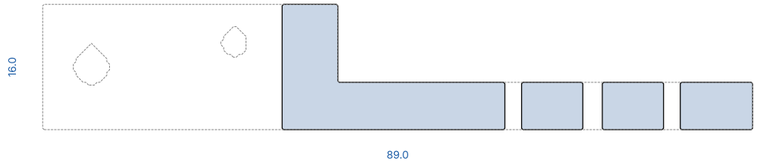
SHEET D14

REV E

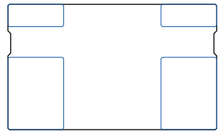
P11 Rudder tiller arm (qty 2)

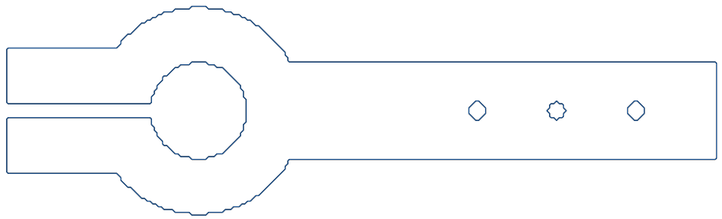
File: stl/P11_tiller_arm.stl Views at 1:0.75 Print orientation: z up from the bed

SECTION y = 0 (front)



SIDE



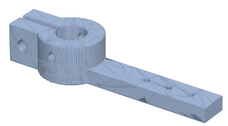


TOP (blue: slice near the bed)

Feature	Size
Hub	D26 × 16
Arm	65 × 12 × 6
Stock bore	D12.1

Material: PETG, 100% infill

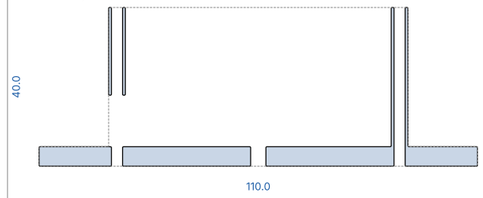
- Rev C: one per rudder (two rudders). Clamps the 12 mm 316 rudder stock (M4 bolt through the ears).
- Cross-drill the stock 3 mm through the D3.2 hole and fit a 316 pin.
- Pushrod holes (D2.2) at 35, 45 and 55 mm from the stock give three throw settings.



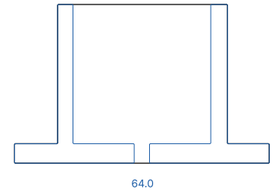
P12 Rudder servo cradle (Blue Trail SER-2000) (qty 2)

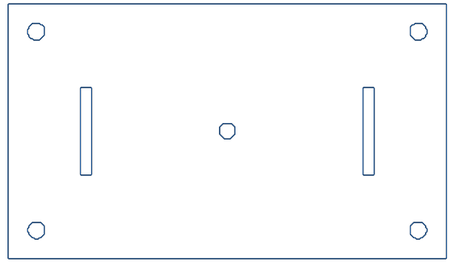
File: stl/P12_rudder_servo_cradle.stl Views at 1:1.5 Print orientation: z up from the bed

SECTION y = 0 (front)



SIDE



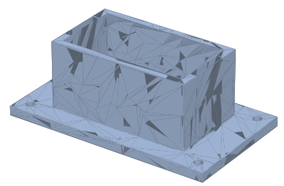


TOP (blue: slice near the bed)

Feature	Size
Base	110 × 64 × 5
Cradle	inside 34.8 wide x 67 long, walls 4, 40 high
Strap slots	2 × 22 × 3

Material: PETG or ASA, 5 walls

- Rev E: one per rudder. Holds a Blue Trail SER-2000 sealed enclosure (with a
- DFRobot magnetic-encoder servo inside) output-up, level with the tiller arm.
- Two 316 hose clamps or 20 mm straps through the slots hold the enclosure;
- no fixed hole pattern, so it fits whatever the unit measures.
- The envelope (66 × 34 × 72) is an ESTIMATE: measure your unit, set SER in
- design.py and rebuild before printing. Base bonds to the pocket pad.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

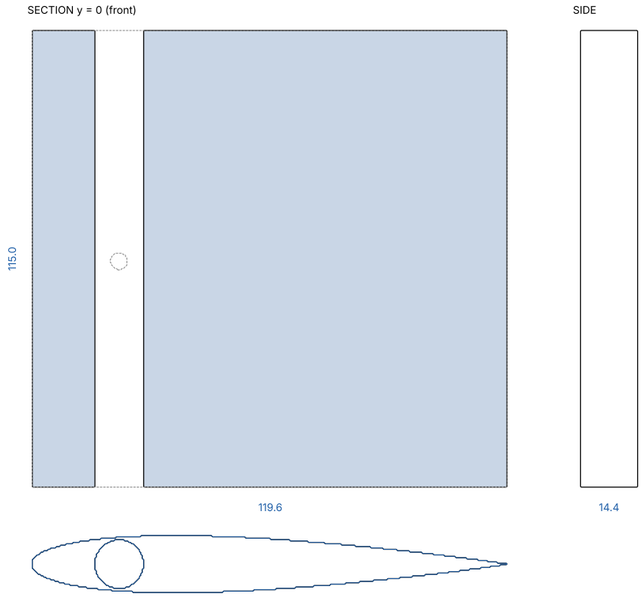
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D15

REV E

P13 Rudder blade core, upper and lower halves (qty 2) [rudder core upper]

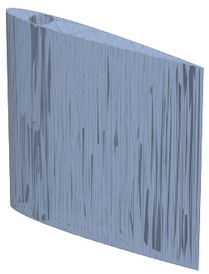
File: stl/P13a_rudder_core_upper.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Chord	120
Span	2 × 115
Stock bore	D12.2 at 22 from LE

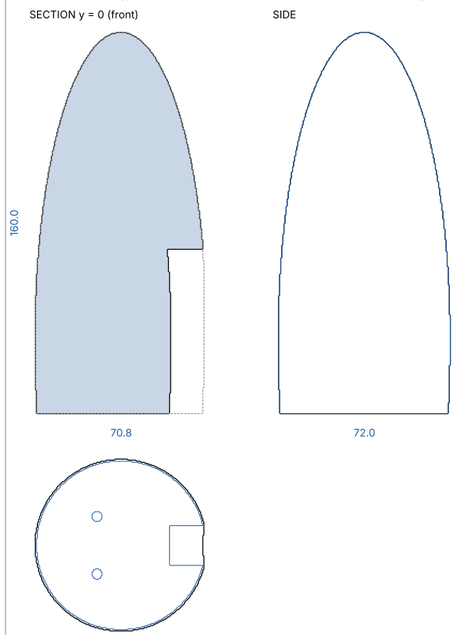
Material: PETG, 15% gyroid

- Rev C: two blades (print two of each half). NACA 0012, 120 mm chord; each half 115 mm, printed standing up.
- Slide both halves on the 12 mm 316 stock, pin each with a 4 mm 316 pin (D4.2 holes), fill the joint with epoxy.
- Sand, then laminate 2 layers of 6 oz glass over the whole blade.
- Stock sits 22 mm behind the leading edge (about 15% balance area).



P14 Keel bulb casting pattern, forward and aft halves [bulb pattern fwd]

File: stl/P14a_bulb_pattern_fwd.stl Views at 1:2.5 Print orientation: z up from the bed



Feature	Size
Length	320
Max diameter	72
Fin slot	150 × 18 × 15

Material: PLA or PETG, 3 walls

- Pattern for a plaster or sand mould: do NOT pour lead into the print.
- Halves split at 160 mm; join with two 4 mm dowels (D4.2 holes) and fill/sand smooth.
- Slot on top matches the fin tip (NACA 0012, 150 mm chord), 15 mm deep.
- Rev C: 72 mm diameter (was 68). Finished lead bulb about 9.2 kg; allow 1-2% for shrinkage when sizing the mould.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

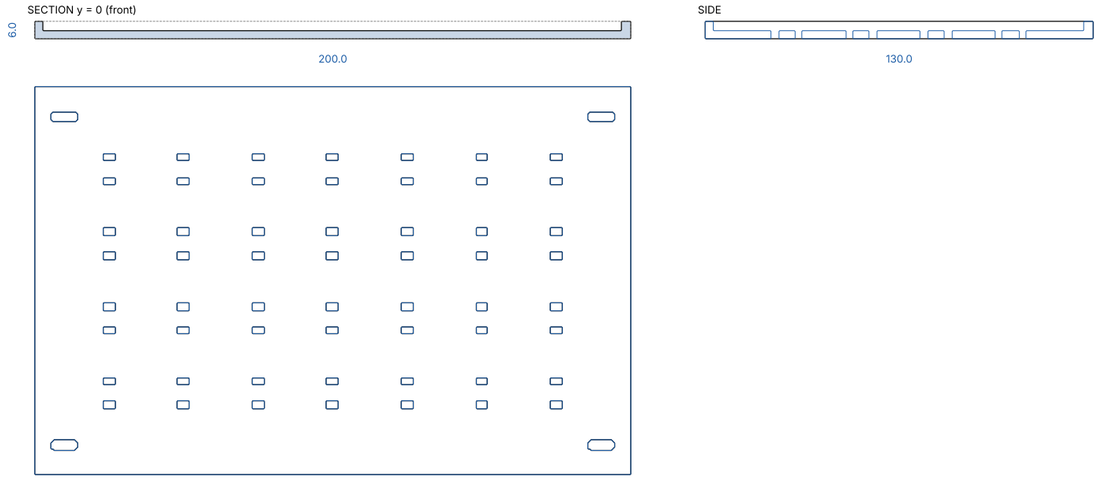
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D16

REV E

P15 Electronics tray

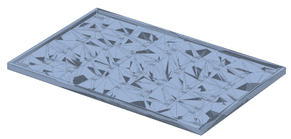
File: stl/P15_electronics_tray.stl Views at 1:2 Print orientation: z up from the bed



Feature	Size
Plate	200 × 130 × 3
Rim	3 high
Zip slots	4 × 2.5 on a 25 mm grid

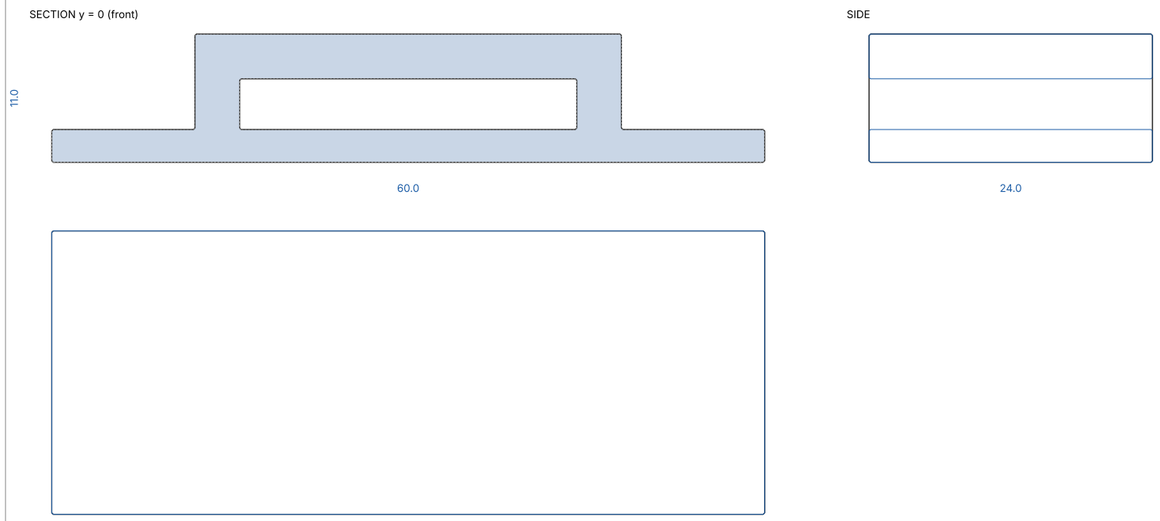
Material: PETG

- Sits inside the IP68 enclosure; trim the outline to your enclosure's inside size.
- Mount boards with 3M Dual Lock and zip ties through the 4 × 2.5 mm slot pairs (no hole patterns are assumed).
- Corner slots take M3 screws into the enclosure's mounting bosses.



P16 Battery strap anchor (qty 4)

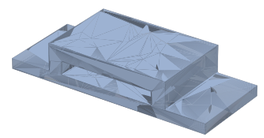
File: stl/P16_battery_strap_anchor_x4.stl Views at 1:0.5 Print orientation: z up from the bed



Feature	Size
Base	60 × 24 × 3
Webbing slot	28 × 4

Material: PETG

- Rev C: bond to the hull floor around the sealed battery box; the straps hold the box (the battery is wedged inside it).
- Use two straps with cam buckles; strap tension holds the battery against capsize loads.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

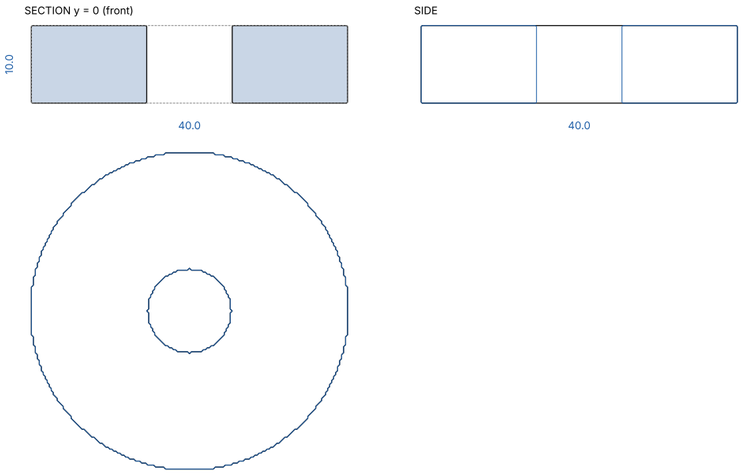
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D17

REV E

P17 Penetrator boss pad (qty 4)

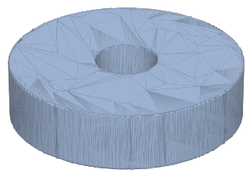
File: stl/P17_penetrator_pad_x4.stl Views at 1:0.75 Print orientation: z up from the bed



Feature	Size
OD	40
Height	10
Hole	D10.5

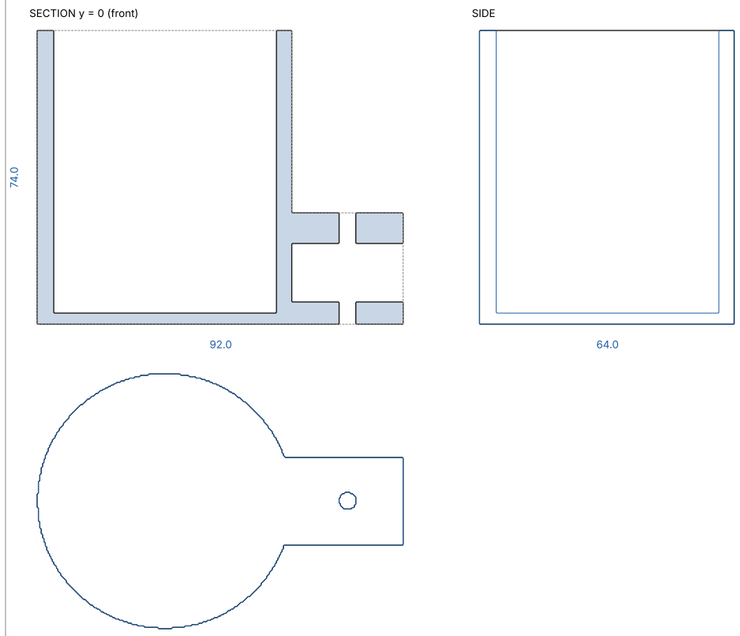
Material: ASA or PETG, 100% infill

- Gives each Blue Robotics WetLink M10 penetrator a flat, thick seat on the curved deck or hull.
- Bond with thickened epoxy, drill through the hull using the D10.5 hole as a guide.



P18 Wing counterweight pod

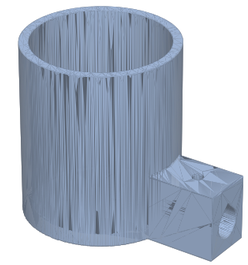
File: stl/P18_counterweight_pod.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Body	OD64 × 74
Cavity	D56 × 71 open top
Arm socket	D12.2 × 28 blind
Socket axis	12 above base

Material: PETG or ASA, 4 walls, 40% infill

- Balances the rotating wing, tail and boom on the spar axis (Rev B; unchanged in Rev C).
- Sits on the forward end of the 12 mm arm (Rev C3: solid fibreglass rod); pod axis 300 mm forward of the spar axis. Pin with M4 (D4.3 hole).
- Fill the open top with lead shot + epoxy (about 0.8 kg, roughly 115 cm3) a little at a time,
- until the assembled rig balances level on the spar with the tail at 0 deg. Cap the top with epoxy.
- Capacity about 175 cm3 (about 1.2 kg) leaves room if the tail parts come out heavier.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

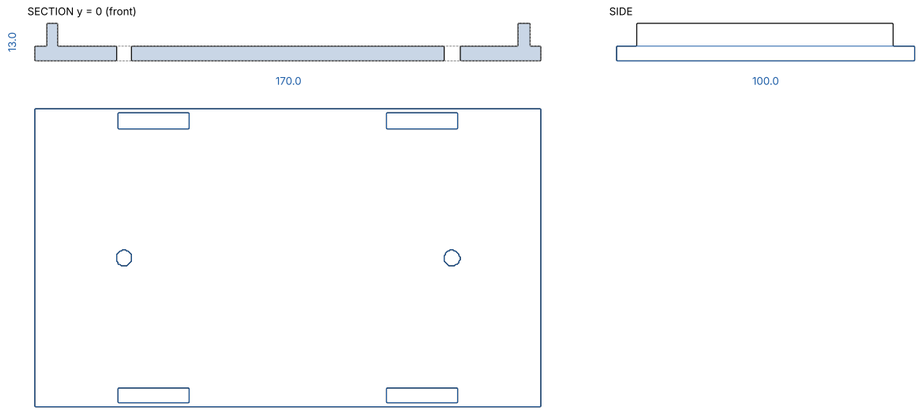
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D18

REV E

P19 Backup tracker (YB3) deck cradle

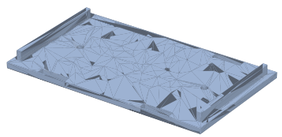
File: stl/P19_yb3_platform.stl Views at 1:2 Print orientation: z up from the bed



Feature	Size
Plate	170 × 100 × 5
End lips	2 × 86 × 4 × 8 high
Strap slots	4 × 24 × 5
Bolt holes	2 x D5.3

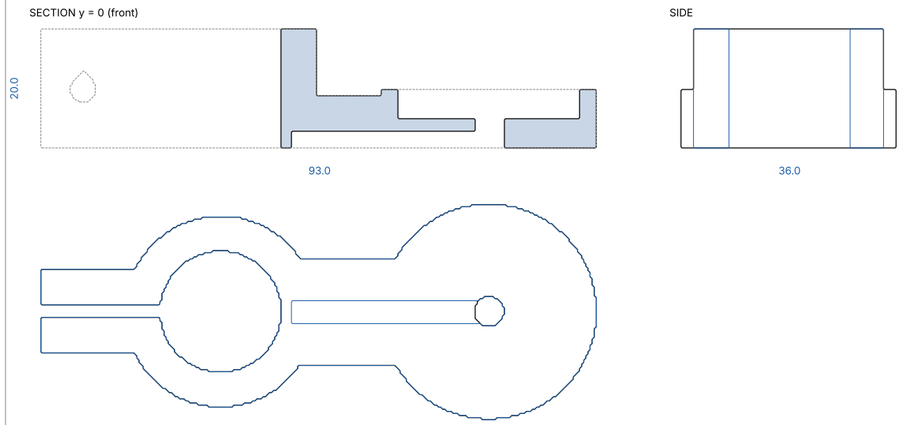
Material: ASA, 5 walls, 40% infill

- Rev C3: the YB3 sits in a 38 mm deep well in the foredeck, centred at x 180, not on the stern post, so a bent post cannot take the backup trac
- Bond the plate to the floor of the well with thickened epoxy and add two M5 316 bolts (D5.3 holes, countersink by hand) with sealant and bac
- Strap the YB3 down with two 316 hose clamps or 20 mm webbing through the four slots; the end lips stop it sliding.
- The YB3 size here is approximate: check your unit before printing.



P20 Navigation light clamp

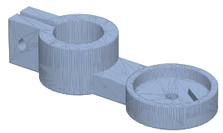
File: stl/P20_nav_light_clamp.stl Views at 1:1 Print orientation: z up from the bed



Feature	Size
Collar	OD32, ID20.3 × 20
Puck seat	D30.5 × 5 deep
Reach	45

Material: ASA

- Rev C: clamps the stub at z 600, below the vane; holds a 30 mm all-round white LED puck 45 mm aft of the stub.
- Cable runs down the D5 hole and along the groove into the stub through a 6 mm hole drilled in the tube.
- The controller switches the light on at night (about 0.3 W). Squeeze slot closed with an M4 bolt.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

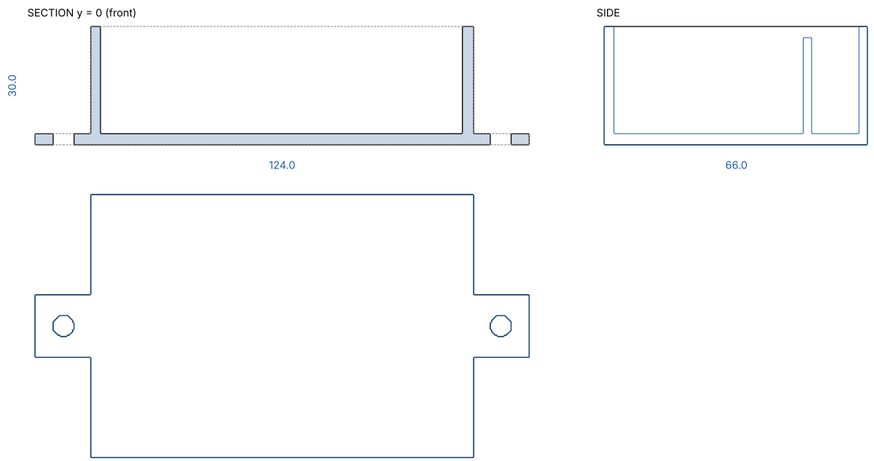
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET **D19**

REV **E**

P21 Limp-home controller case (potting shell)

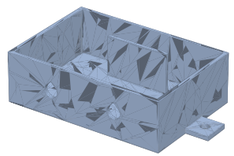
File: stl/P21_limp_home_case.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Case	96 × 66 × 30
Walls	2.5 (top 2.0)
GPS shelf	38 wide, 12 under the top
Cable holes	2 x D8
Ears	2 x D5.3

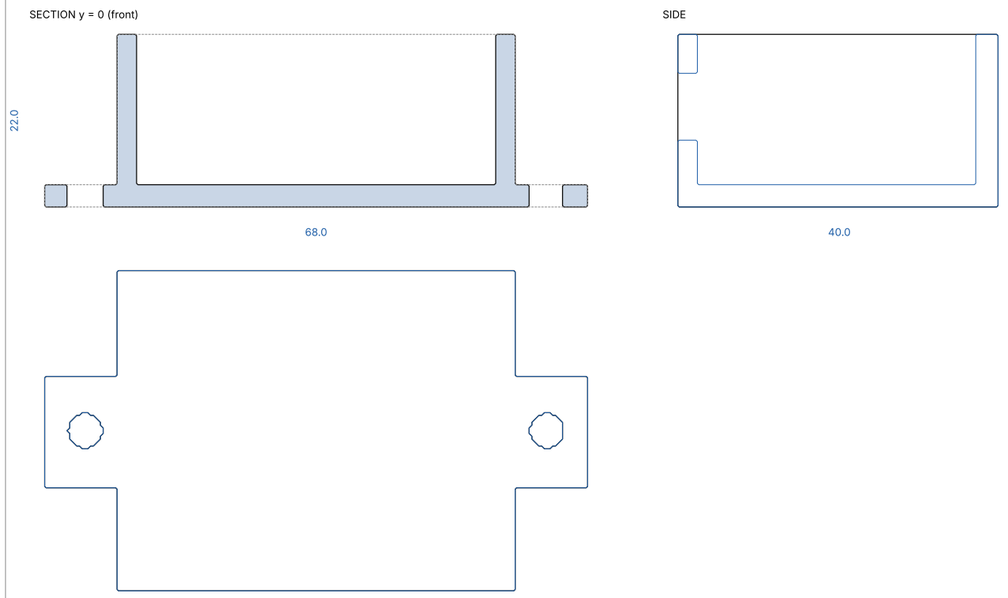
Material: PETG, 4 walls, 100% infill on the back

- Rev D (docs/Limp_Home_Spec.md). Bond the back to the aft face of
- bulkhead x 300, top edge 10 mm under the deck, clear of the hatch opening.
- Print back-down. ESP32, 6 V regulator, opto-isolators and supervisor
- on the floor; GPS and BNO085 face-up on the shelf under the top wall.
- Pot in two pours: soft silicone round the GPS and BNO085, then
- urethane to the rim. Cables out through the D8 holes (drip loop).
- Pot only after Bench Test N. Updates over Wi-Fi: no connector inside.



P22 Servo switchover pod (potting shell)

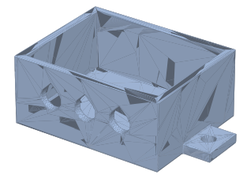
File: stl/P22_switchover_pod.stl Views at 1:0.75 Print orientation: z up from the bed



Feature	Size
Case	50 × 40 × 22
Walls	2.5
Cable holes	3 x D7
Ears	2 x D4.3

Material: PETG

- Rev D: holds the 4-channel servo multiplexer and two ideal-diode
- boards for the starboard rudder and tail servos. Bond to the forward
- face of bulkhead x 1260, high, beside the servo WetLink.
- Pot to the rim with urethane after Bench Test N.
- Cable holes: main box, limp-home controller, servos.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

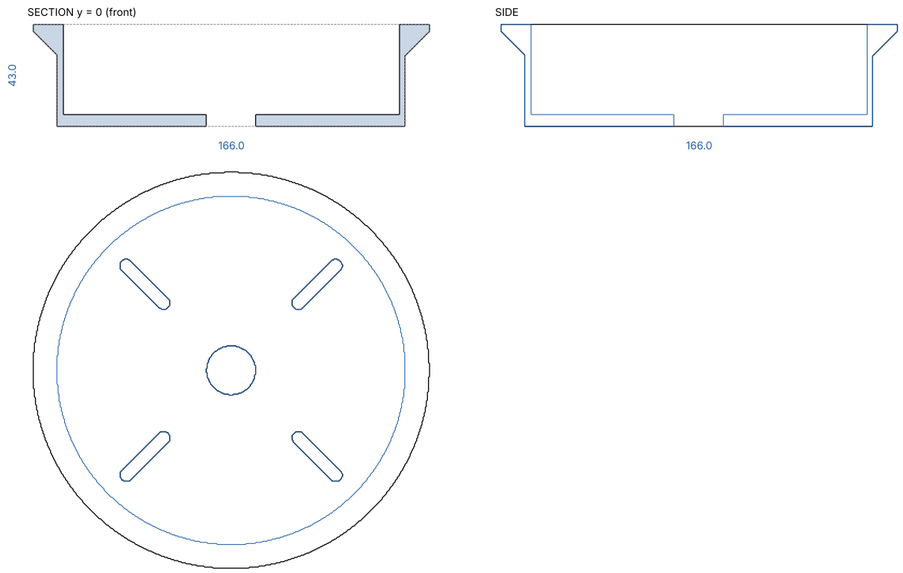
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D20

REV E

P23 RockBLOCK Plus bowl (hatch lid)

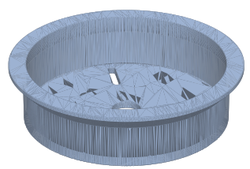
File: stl/P23_rockblock_plus_bowl.stl Views at 1:2.5 Print orientation: z up from the bed



Feature	Size
Flange	OD166 × 3, 45 deg chamfer under
Bowl	ID141 × 38 deep, wall 2.5
Floor	5 thick, D20.5 gland hole
Bolt slots	4 × 5.5 wide, r 40-62

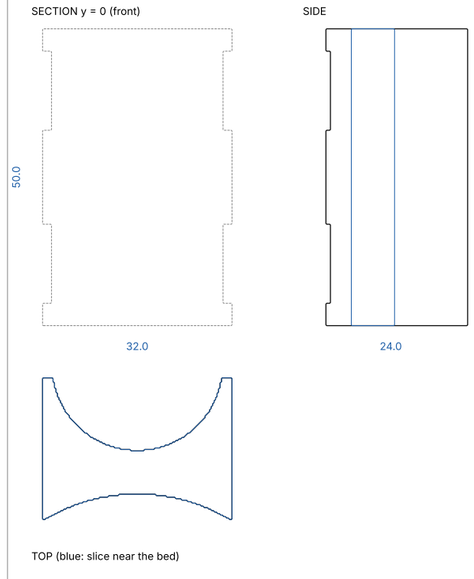
Material: ASA, 5 walls, 40% infill

- Rev D: seats the RockBLOCK Plus (137 × 40) in the hatch lid, its
- top 2 mm proud of the flange and below z 255 (counterweight pod sweep).
- Cut D146 in the lid (T16); bed the flange in thickened epoxy, then one
- 6 oz tape over the flange onto the lid. Print floor-down: the flange
- is chamfered underneath, so no supports.
- Bolt the puck through the four slots into its M5 inserts (check the
- pattern on your unit). M20 IP68 cable gland in the centre hole.
- Soft sealant fillet round the top of the puck: no standing water.



P24 Bilge pump spacer (mast tube)

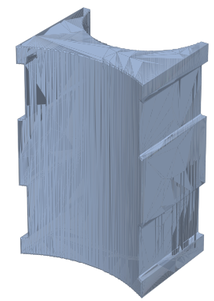
File: stl/P24_bilge_pump_spacer.stl Views at 1:1 Print orientation: z up from the bed



Feature	Size
Faces	R30.4 (tube OD 60.3) and R14.3 (pump D28)
Size	32 × 26 × 50
Clamp grooves	2 × 13 wide x 1.5 deep

Material: PETG, 5 walls, 40% infill

- Rev E: holds the 12 V diaphragm pump (D28 motor) upright against the port
- side of the 2 in. PVC mast tube, above the sump. Bond with thickened epoxy
- (scuff the PVC), then a 316 band clamp round tube, spacer and pump at the
- top and bottom of the spacer (grooves). Pump head down, inlet at the bottom.
- Pick-up hose to a strainer at the sump's low point (x 600, centreline);
- outlet up to a loop under the deck, check valve, deck fitting (D03, D08).
- Print standing on end as modelled: no supports.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

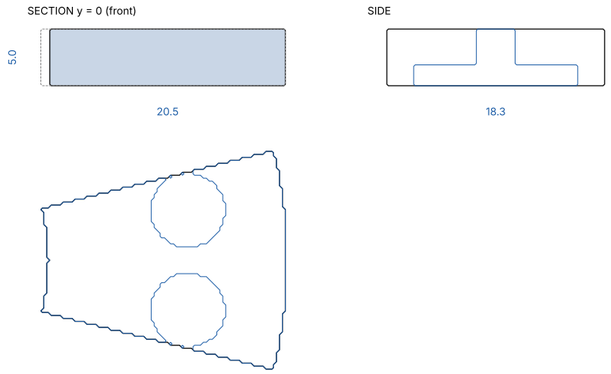
Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D21

REV E

P25 Tail detent magnet carrier

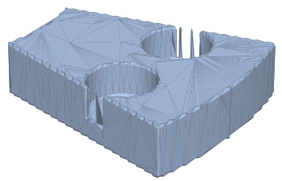
File: stl/P25_tail_detent.stl Views at 1:0.5 Print orientation: z up from the bed



Feature	Size
Sector	r 18-38, +/-14 deg, 5 thick
Magnet pockets	2 x D6.2 × 3.2 at r 30, +/-8 deg

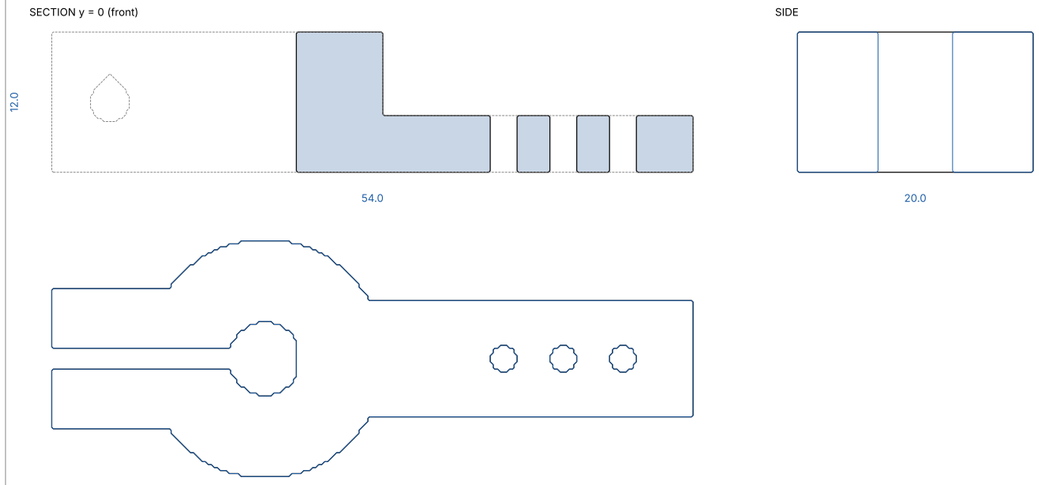
Material: PETG, 100% infill

- Rev E: a tail servo that fails free kills the boat; this makes it fail at
- +/-8 deg. Bond on top of P09, aft of the servo output, centred on it.
- Two 6 × 3 N52 magnets (epoxied, same pole up) in the pockets at r 30, +/-8 deg.
- A 6 × 1 steel disc in the underside of the fin P10, at r 30 on the fin's
- chord line, is pulled to the nearer magnet when the servo lets go.
- Check a 1.5 mm gap to the fin; the powered servo overcomes the pull easily.



P26 Rudder servo arm for the 6 mm D-shaft (qty 2)

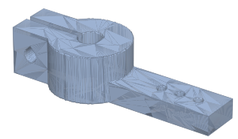
File: stl/P26_rudder_servo_arm.stl Views at 1:0.5 Print orientation: z up from the bed



Feature	Size
Hub	D20 × 12
Bore	D6.1 with flat at 5.5
Arm	36 × 10 × 5

Material: PETG, 100% infill

- Rev E: the SER-2000 output is a 6 mm titanium D-shaft, not a 25T spline,
- so the C3 spline servo-saver is replaced by a spring overload link in the
- pushrod (D05). Clamp with an M3 316 bolt across the slit.
- Pushrod holes D2.2 at 20, 25 and 30 mm to match the tiller's throw.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

Date 2026-11-13 Drawn with Claude Check all dims before cutting

SHEET D22

REV E